

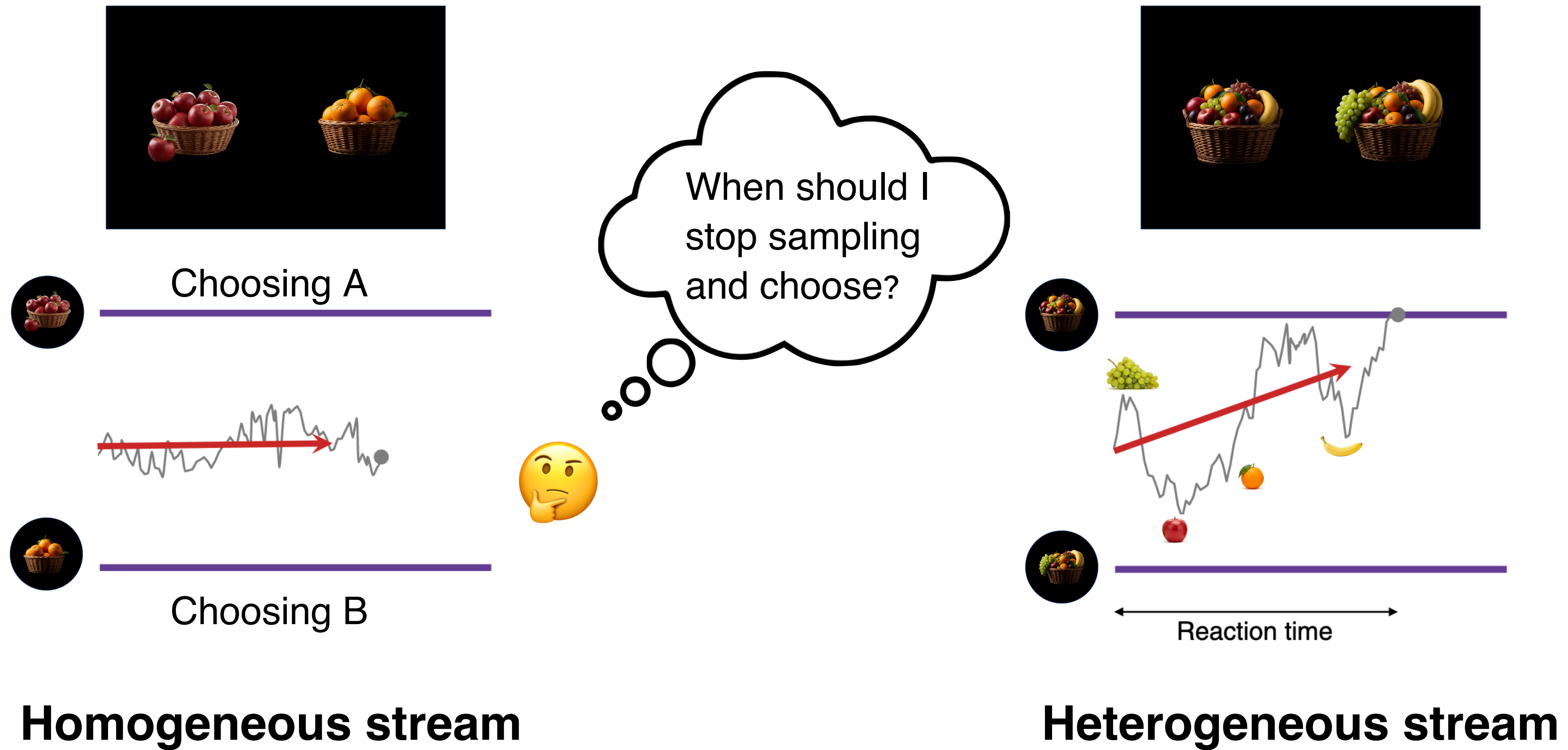
A recurrent network reveals early commitment as a novel strategy in sequential sampling task

Glimcher lab meeting

Jialin Li, Mar 25th

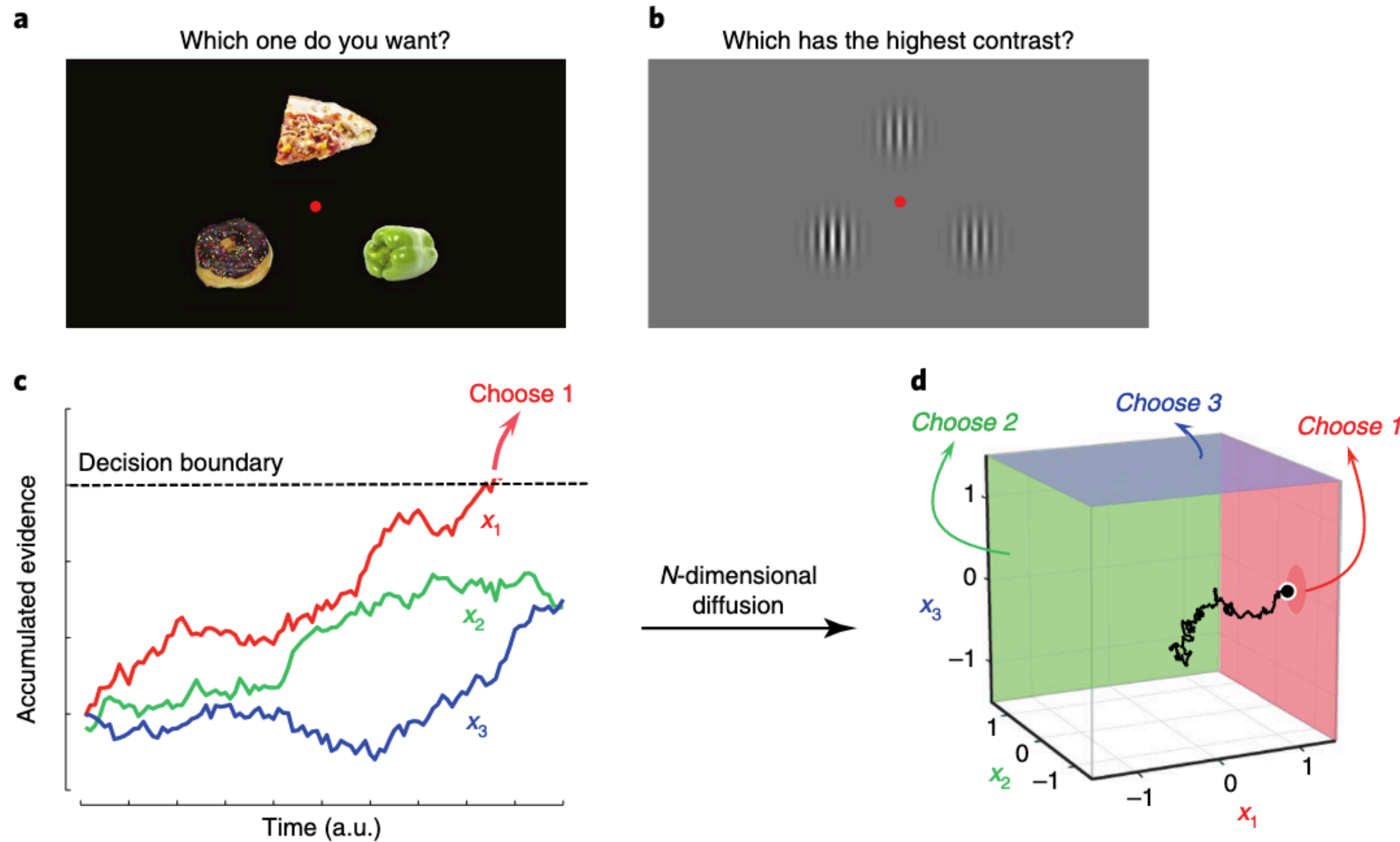
Motivation

Optimal Stopping Rule in Heterogeneous environments



Introduction

Decision making as a dynamic evidence accumulation process



Optimal stopping rule

$$V(t, \mathbf{x}) = \max \left\{ \max_i r_i(t, \mathbf{x}), \left\langle V(t + \delta t, \mathbf{x}(t + \delta t)) \right\rangle - c\delta t \right\}$$

Optimal Evidence accumulation

$$\delta \mathbf{x}_n | \mathbf{z} \sim \mathcal{N}(\mathbf{z}\delta t, \Sigma_x \delta t)$$

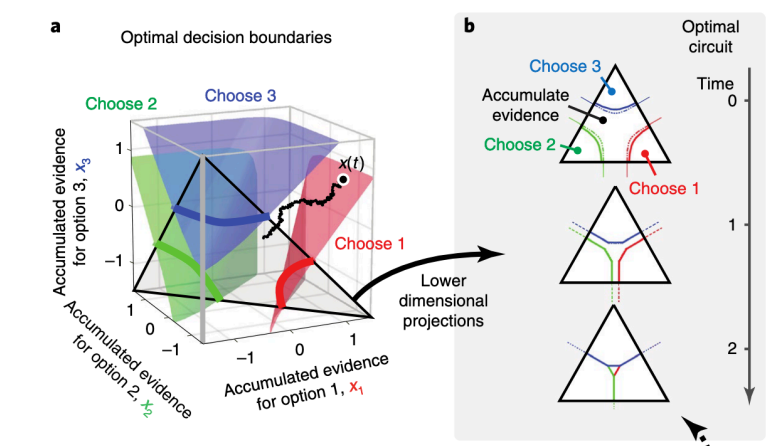
$$\mathbf{x}(t) = \sum_{n=1}^N \delta \mathbf{x}_n, \quad t = N\delta t.$$

$$p(\mathbf{z} | \mathbf{x}(t)) \propto p(\mathbf{x}(t) | \mathbf{z}) p(\mathbf{z}).$$

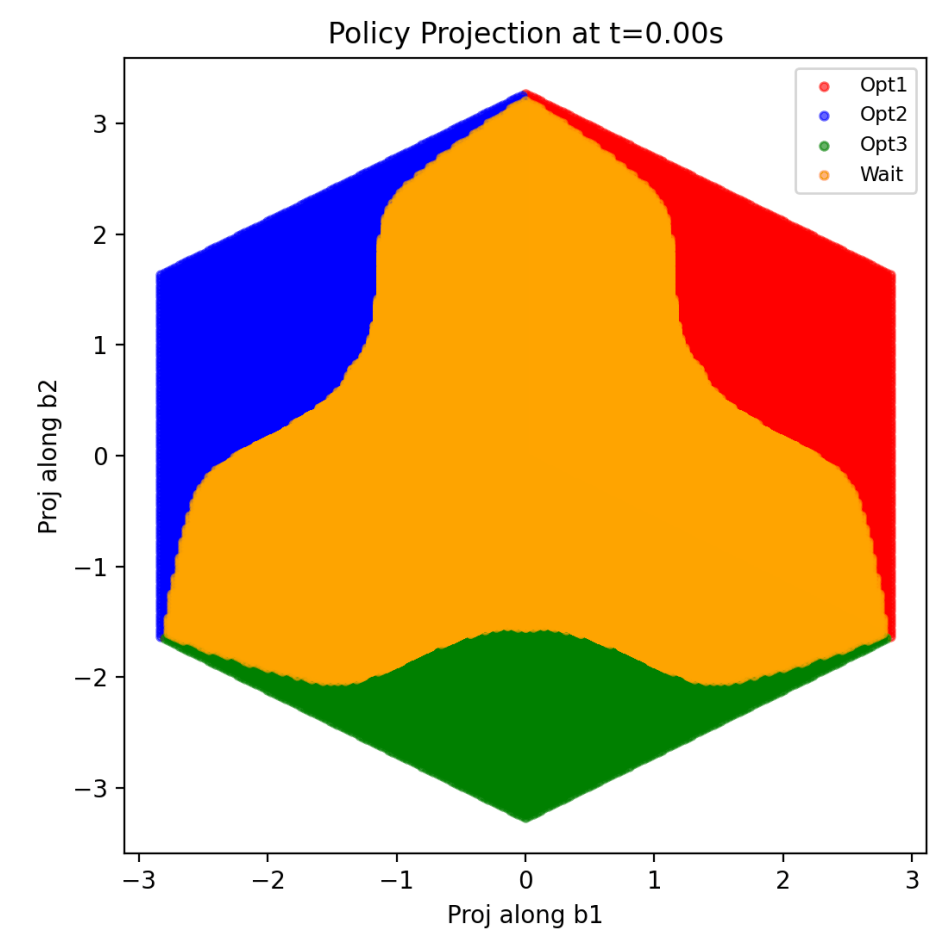
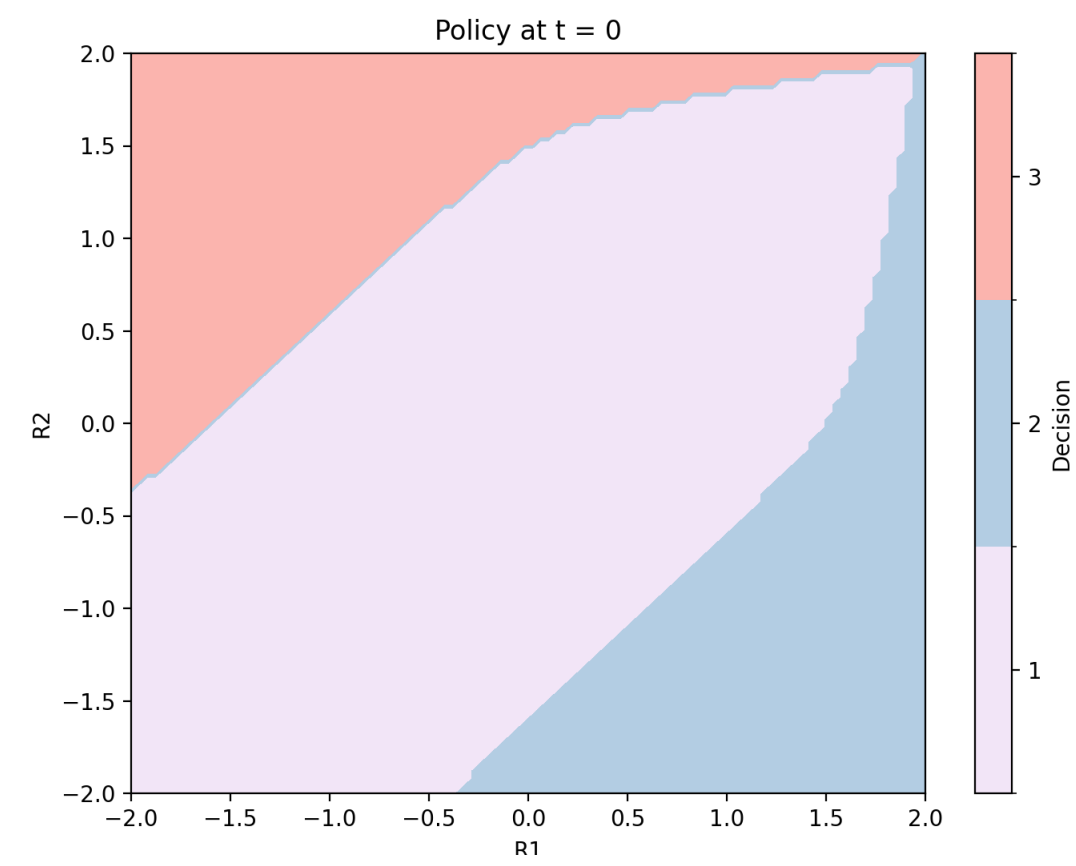
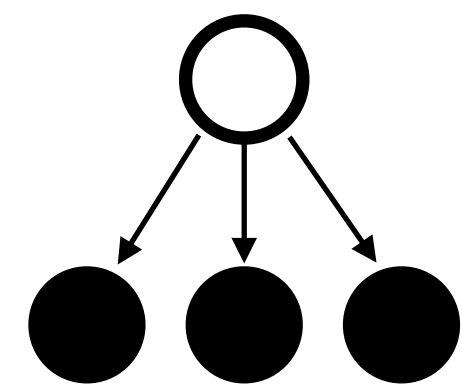
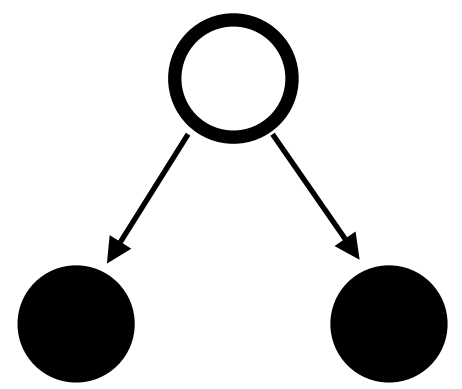
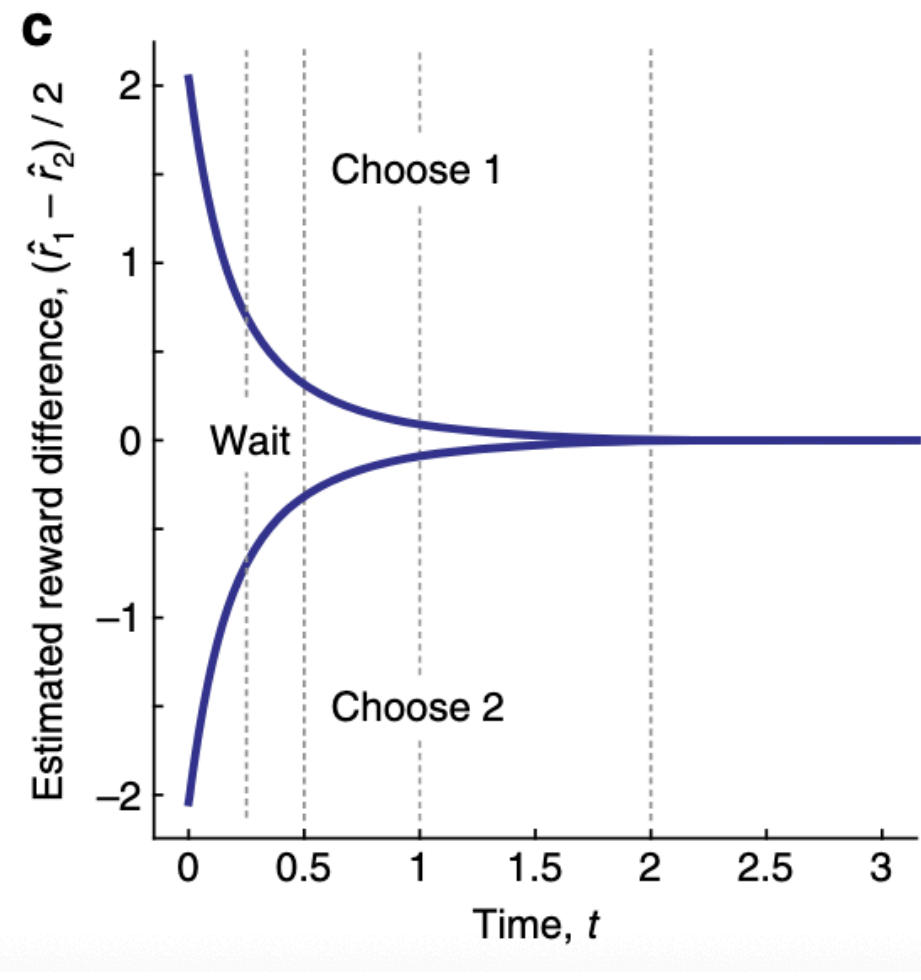
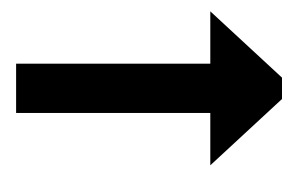
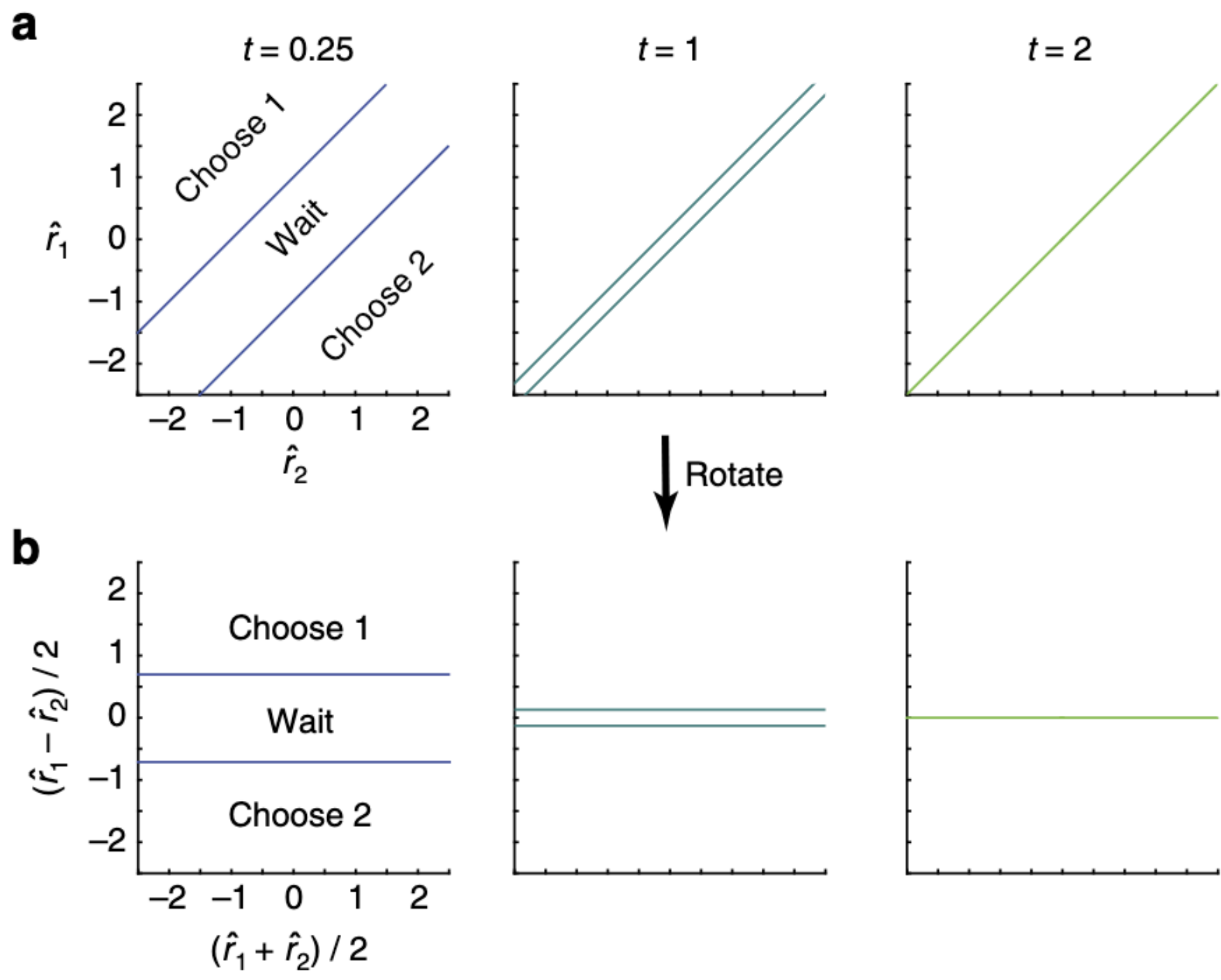
(Tajima et al, 2016; 2019)

Introduction

Optimal policy in simple choice decision making



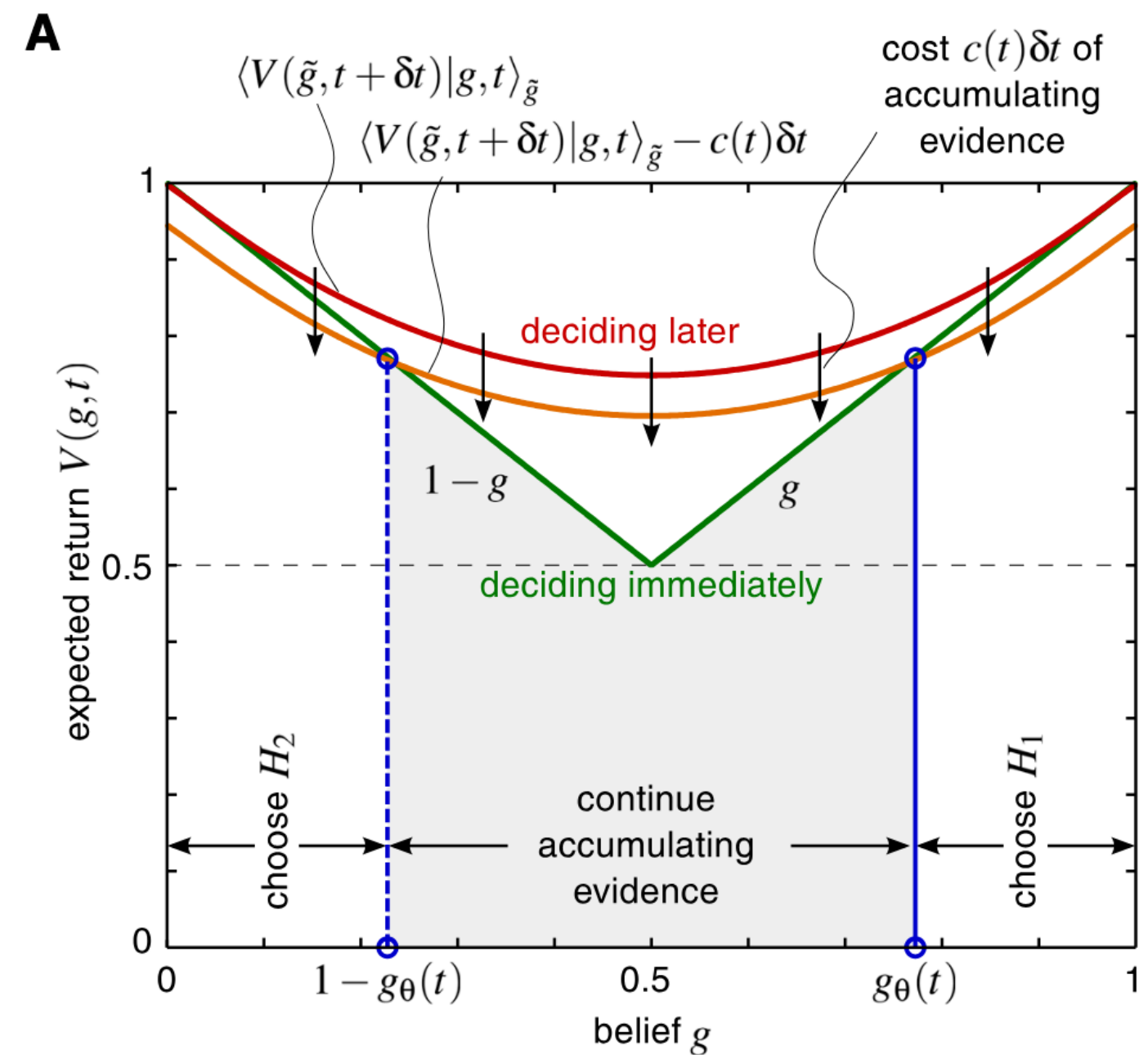
(Tajima et al., 2019)



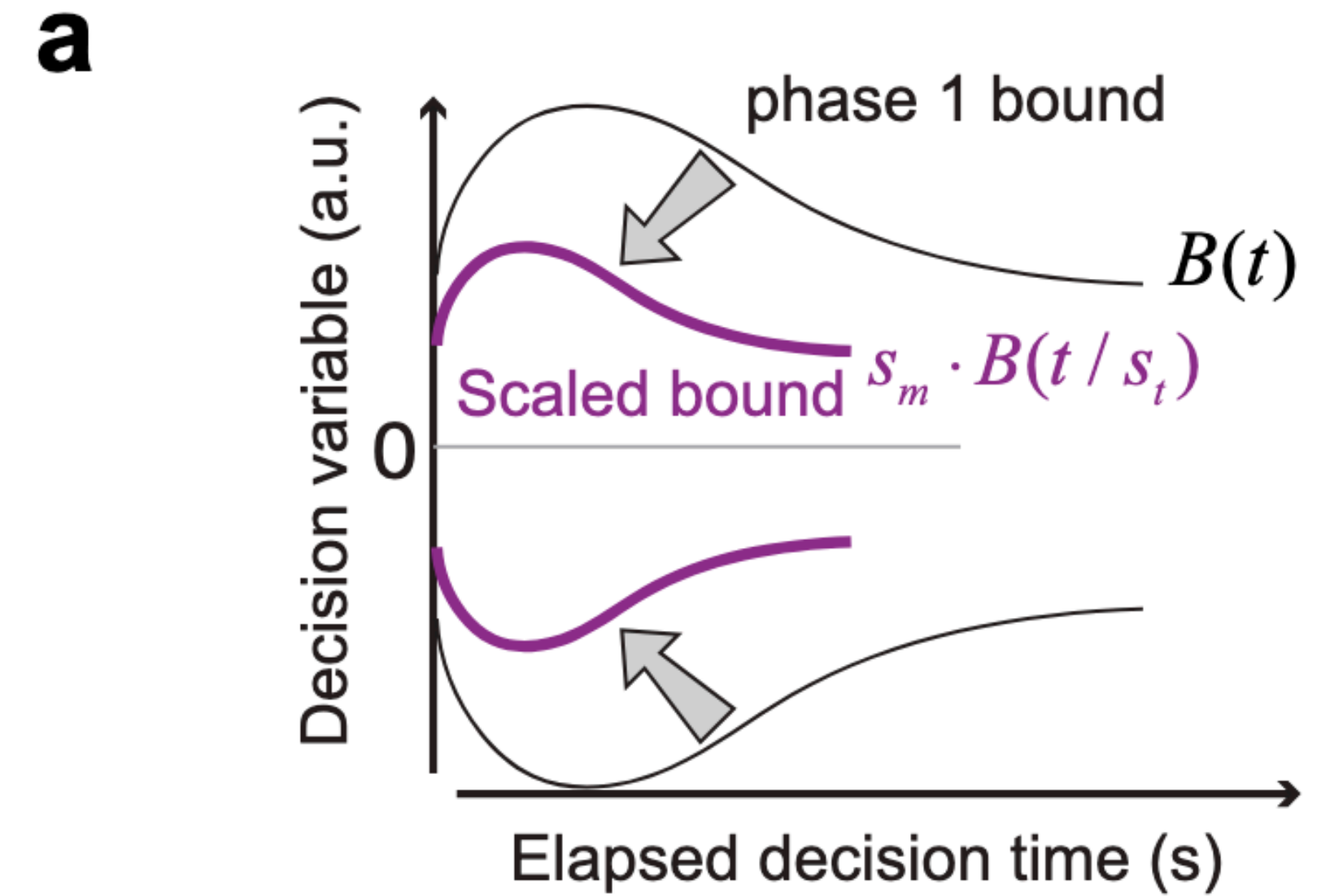
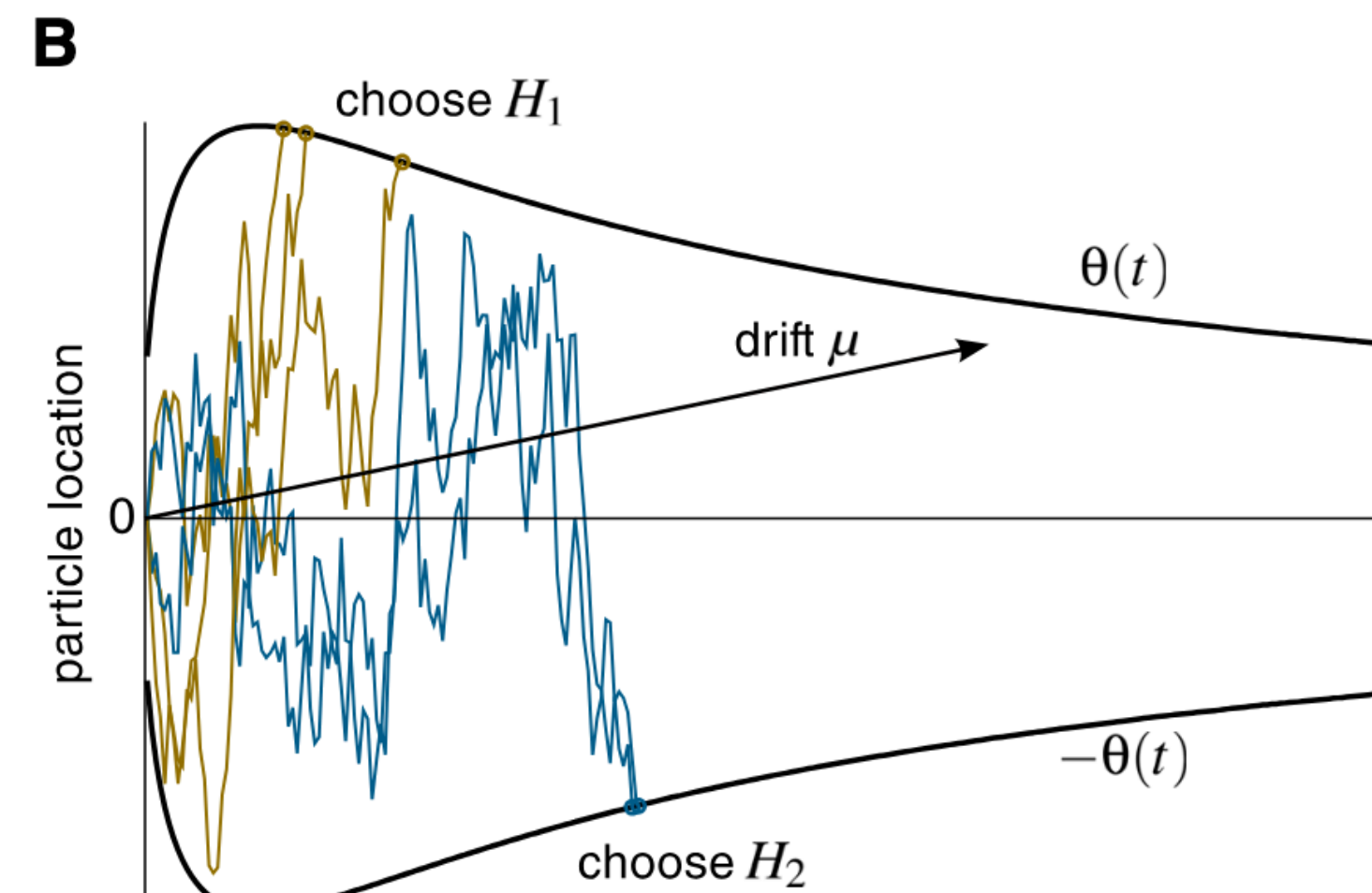
Tajima et al, 2016

Introduction

Time varying decision boundary



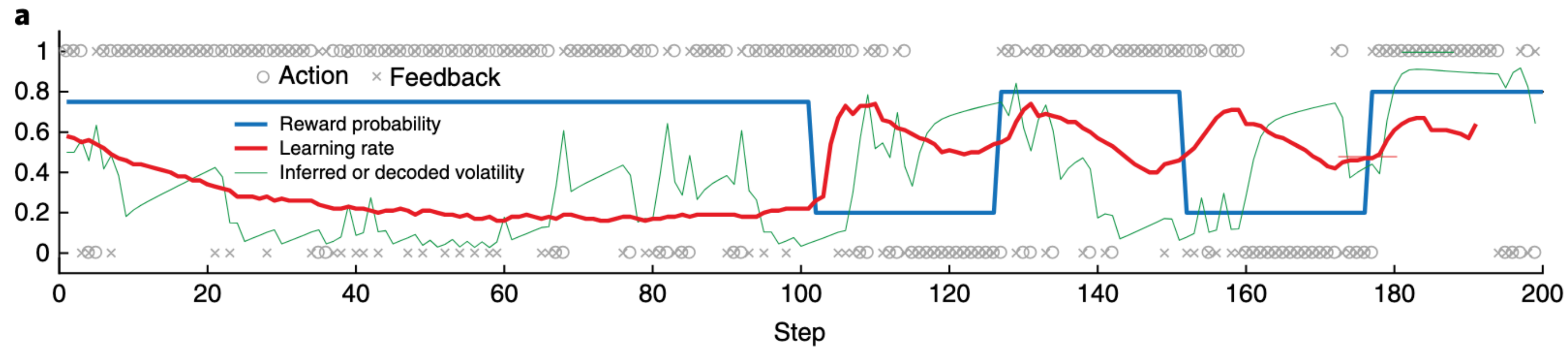
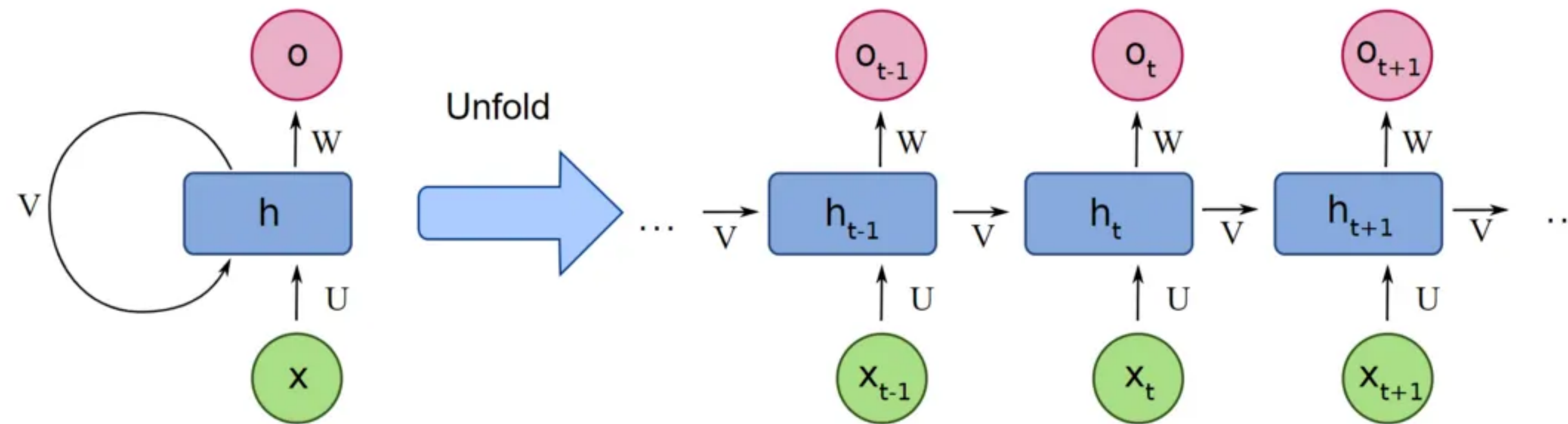
Drugowitsch et al., 2012. Journal of Neuroscience



Kira et al., 2025. Journal of Neuroscience

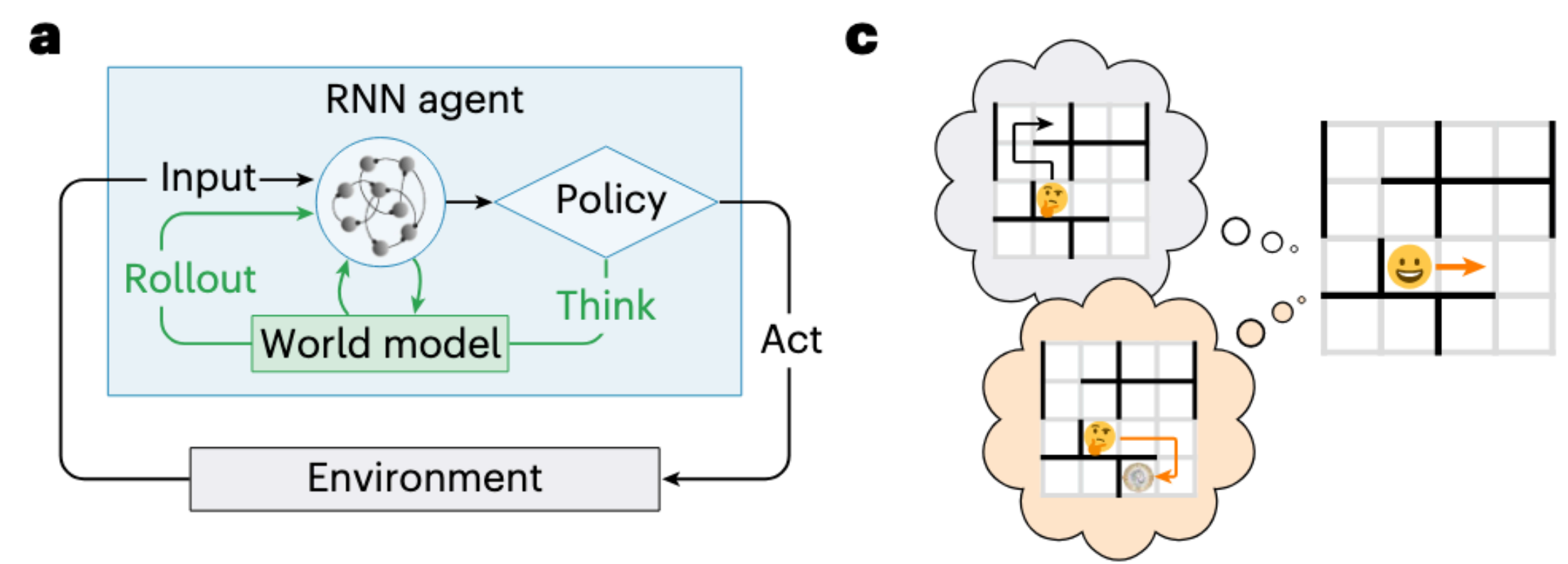
Introduction

Why use RNNs?

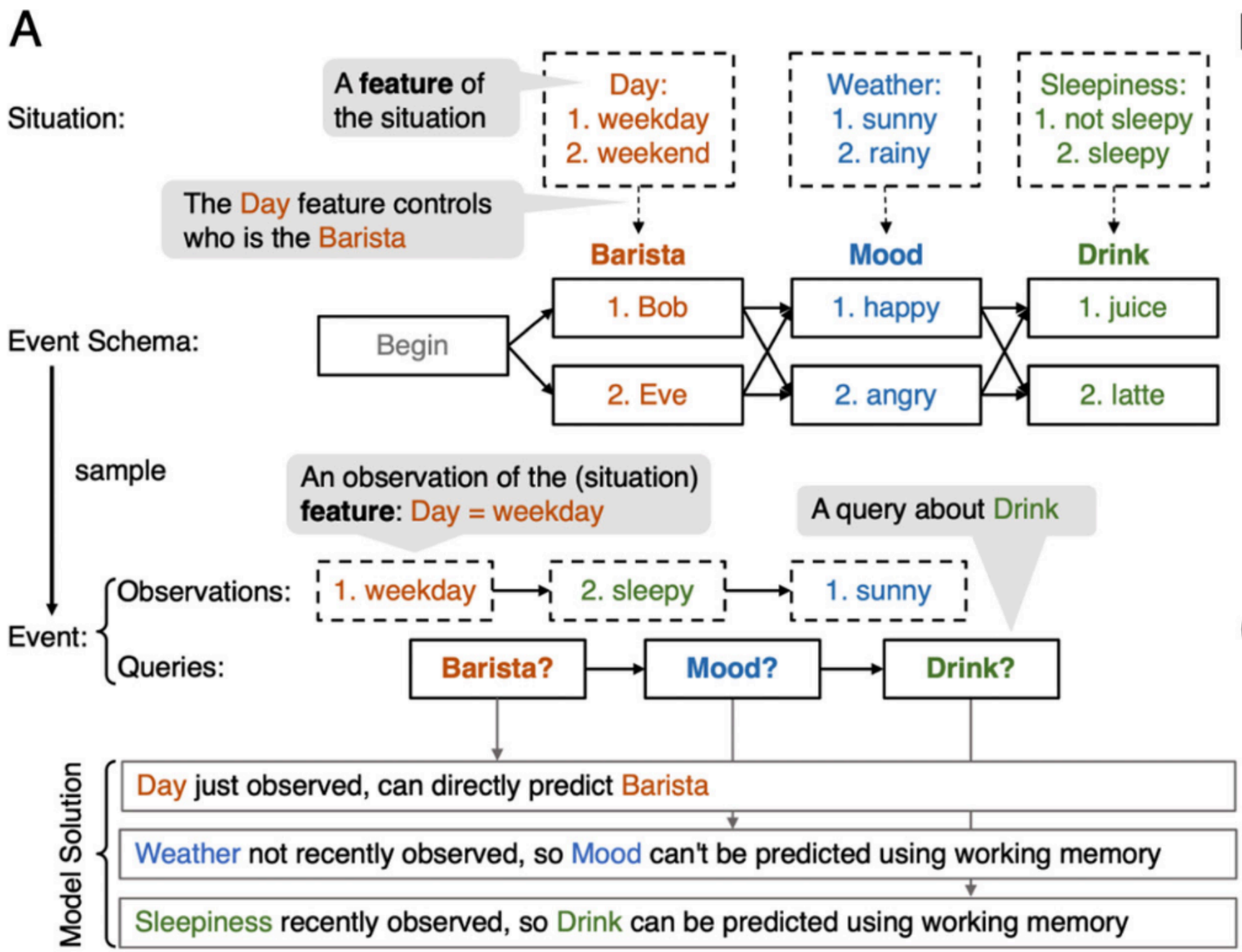


Introduction

Why use RNNs?



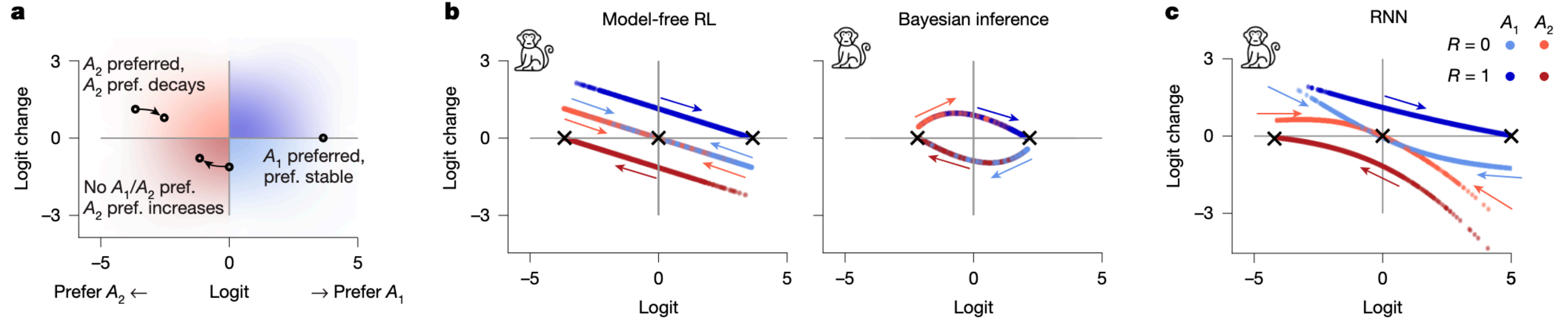
Planning task



Episodic Memory task

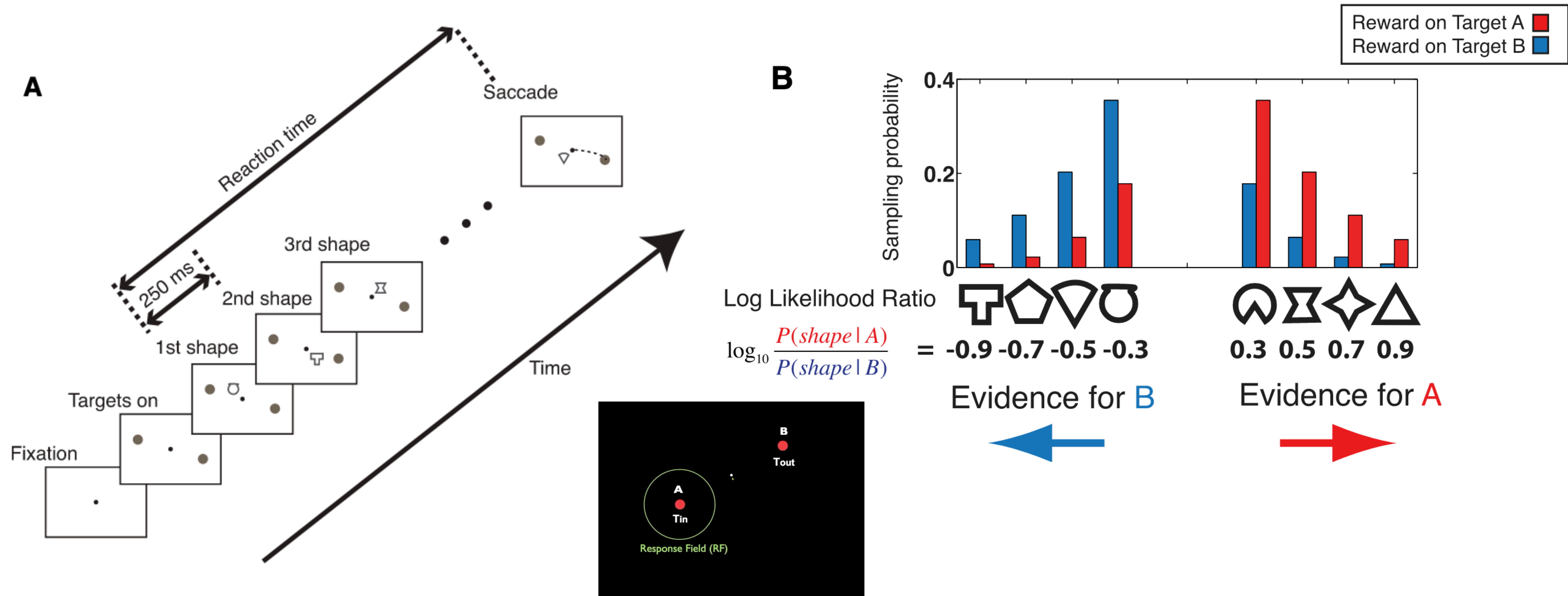
Introduction

Why use RNNs?



Introduction

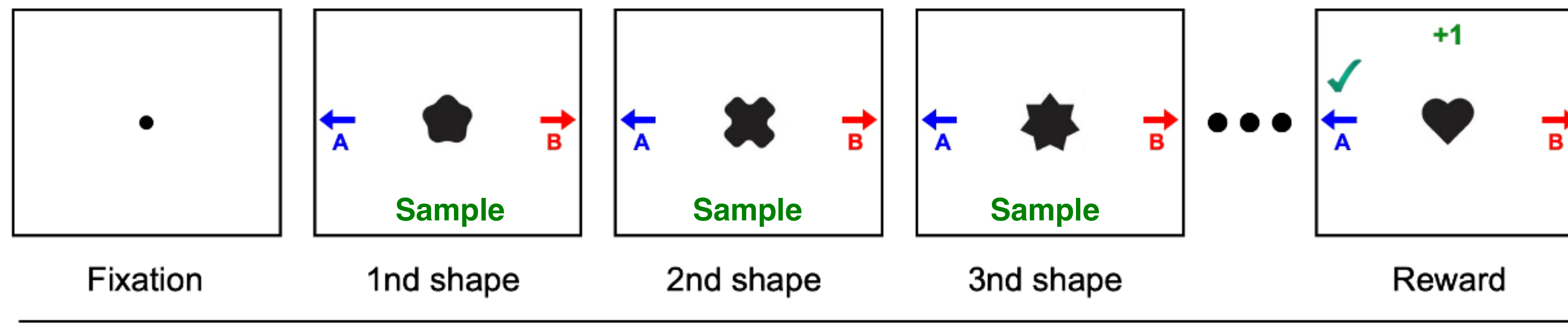
Sequential Probability Ratio Task as sequential evidence accumulation process



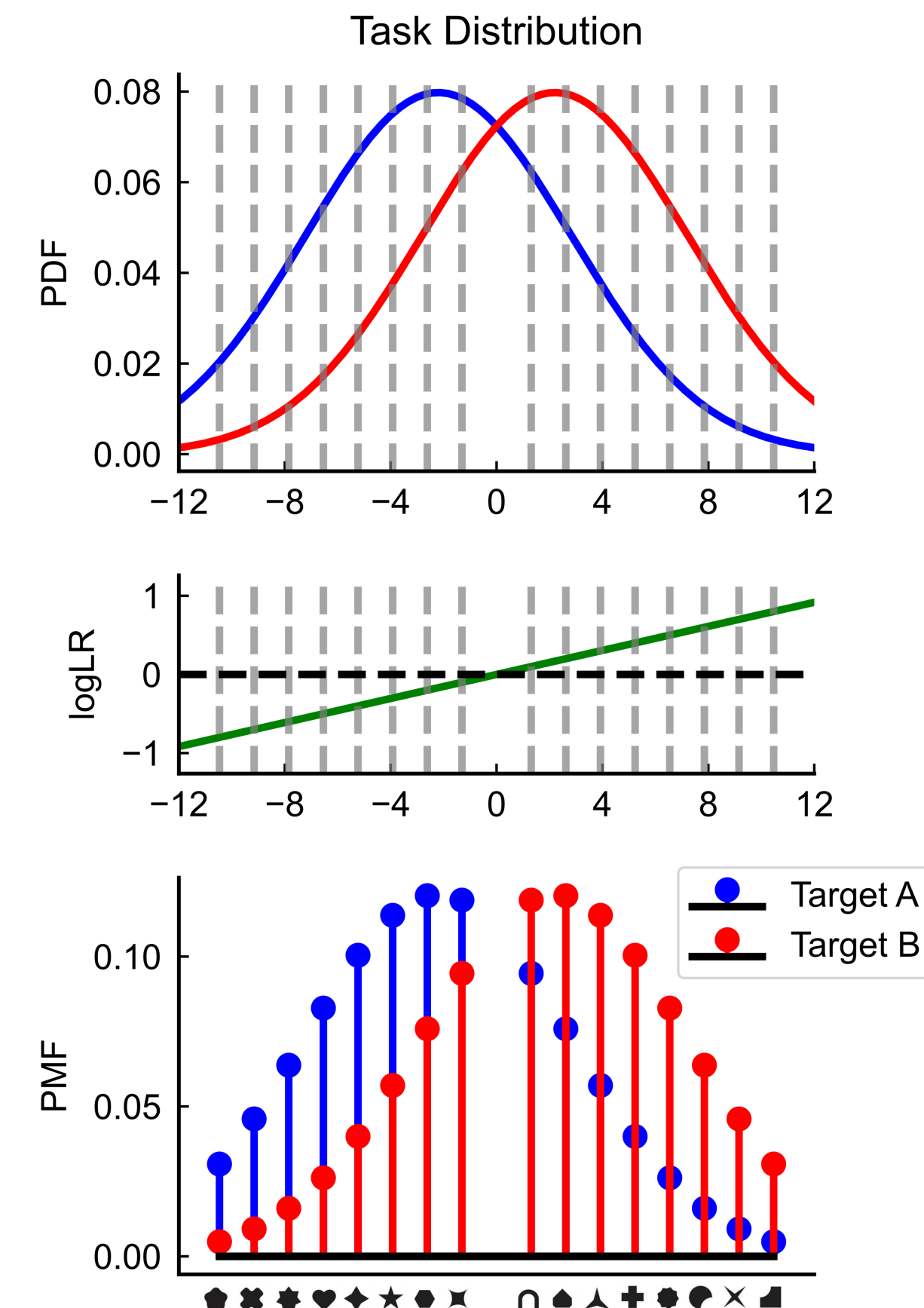
Method

Sequential Sampling Decision-making Task

- **Hidden target (A or B)** randomly selected per trial, determining the sampling frequency of the stimuli.
- Agent receives one stimulus at a time sequentially from the given Gaussian distribution
- Task for the agent: deciding to choose A/B or continue sampling (cost c) at each time.



Heterogeneous stream of evidence over time steps

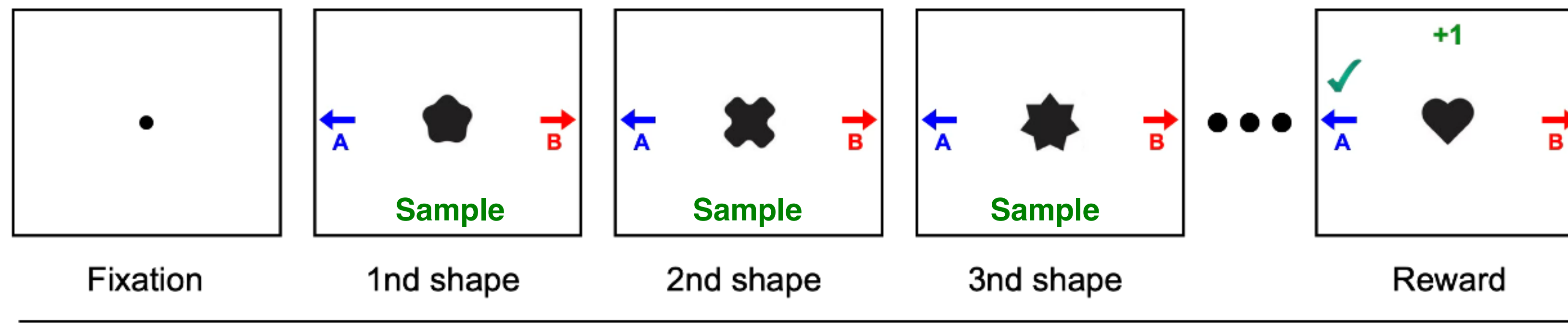


Method

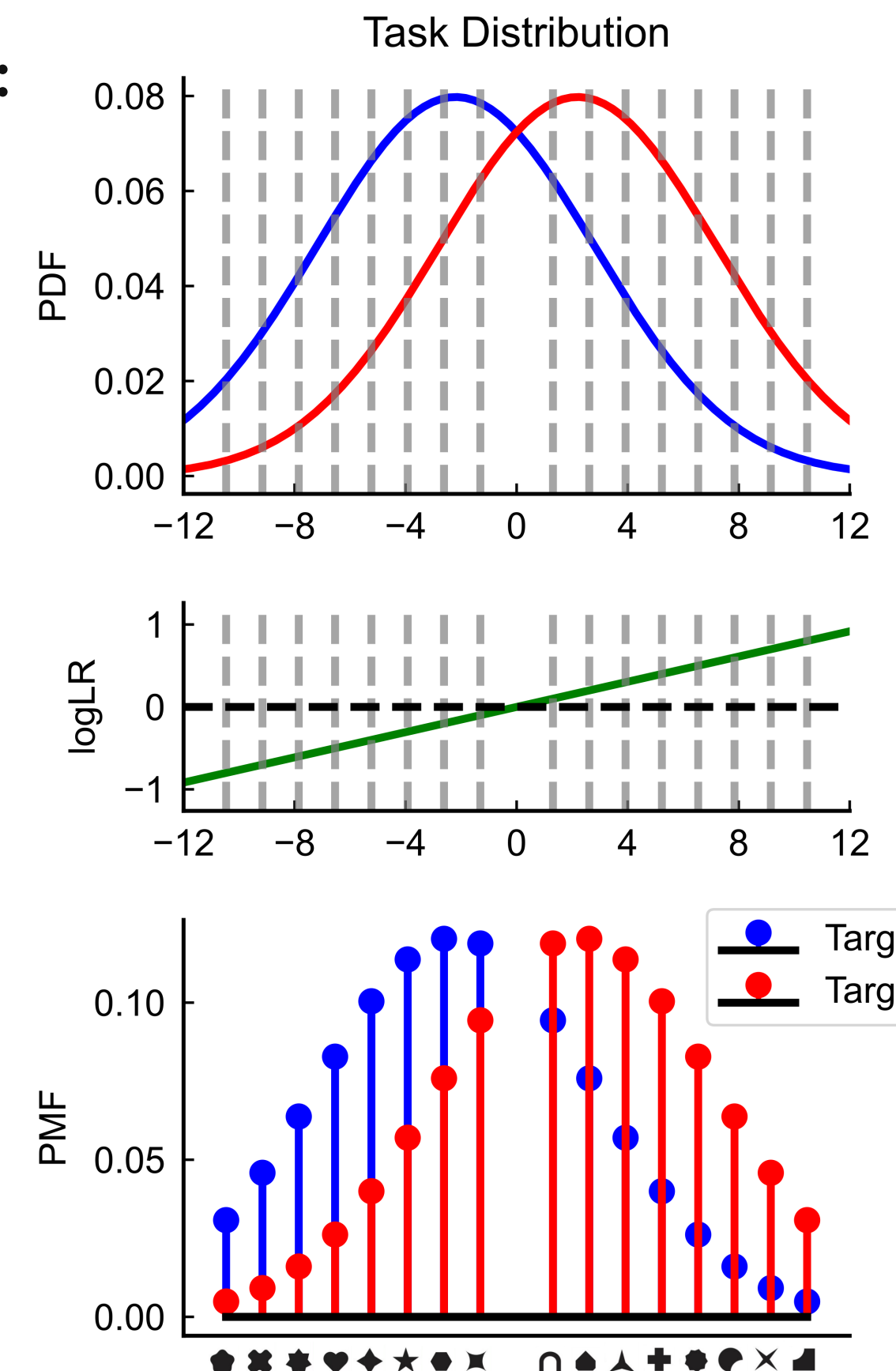
Sequential Sampling Decision-making Task

To study the stopping rule in heterogeneous environment, we tested two factors:

- Sampling cost (5 levels): 0.01, 0.02, 0.03, 0.04, 0.05
- Time constraints (2 levels): No time constraint / time constraint at time step 10



Heterogeneous stream of evidence over time steps

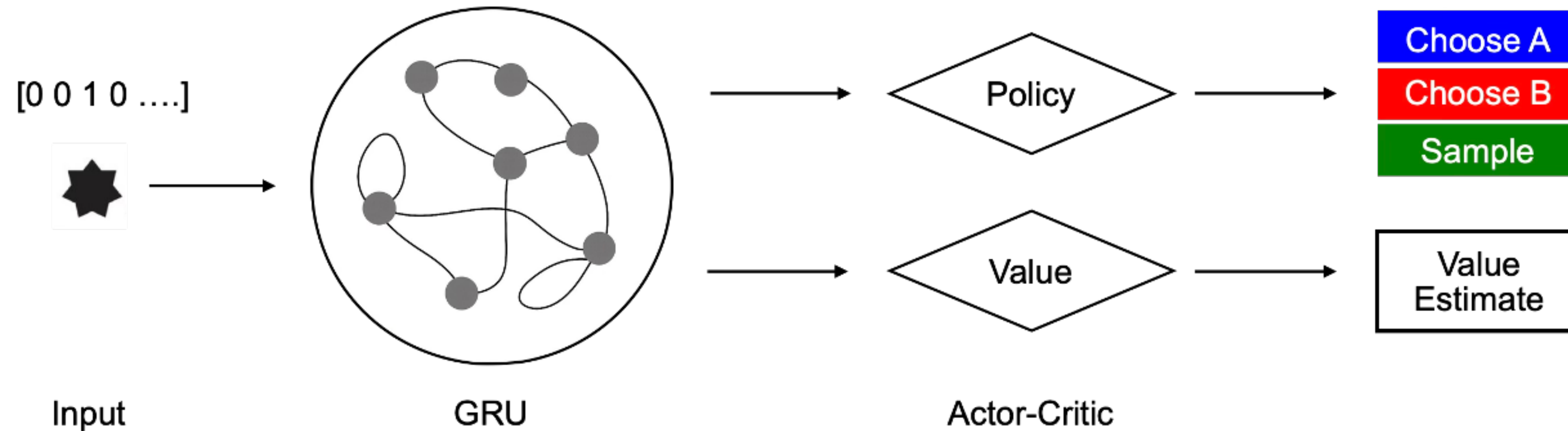


Method

Model Architecture

- The model consists of GRU cell with 64 hidden units, a policy head and a value head.
- The models were trained on-policy using an **advantage actor-critic (A2C) algorithm**:

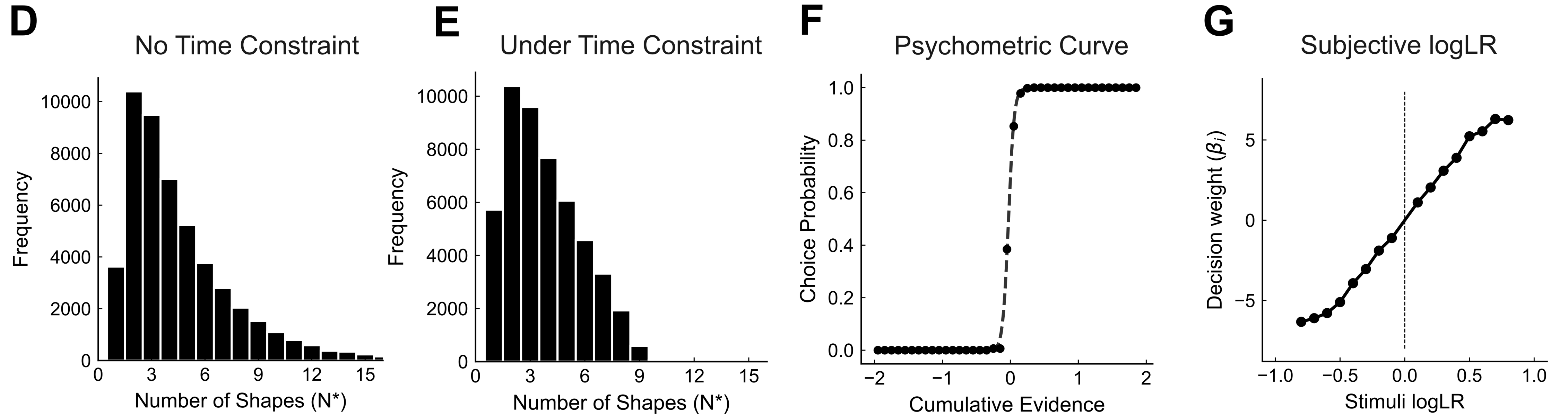
$$J(\theta, \phi) = \mathbb{E}_t \left[\underbrace{\log \pi_{\theta}(\mathbf{a} | s_t) A_t}_{\text{Policy term}} + \underbrace{\beta_v \left(R_t - V_{\phi}(s_t) \right)^2}_{\text{Value term}} + \underbrace{\beta_e \mathbb{H}(\pi(\cdot | s_t))}_{\text{Entropy term}} \right]$$



Results

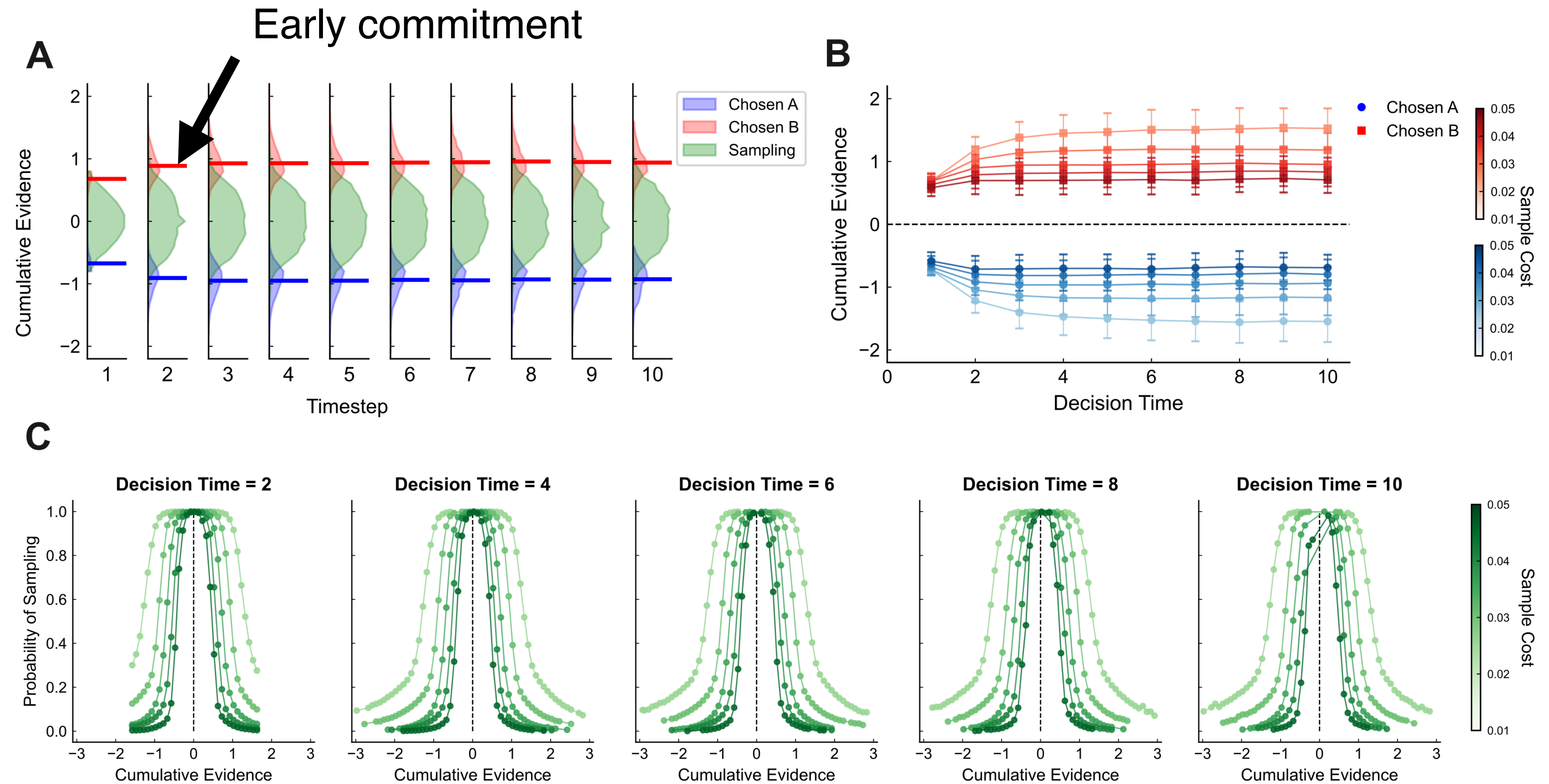
Psychometric Curves under different time constraint conditions

$$\text{logit}(P(\text{chosen B})) = \beta_0 + \sum_{i=1}^{16} \beta_i n_i$$



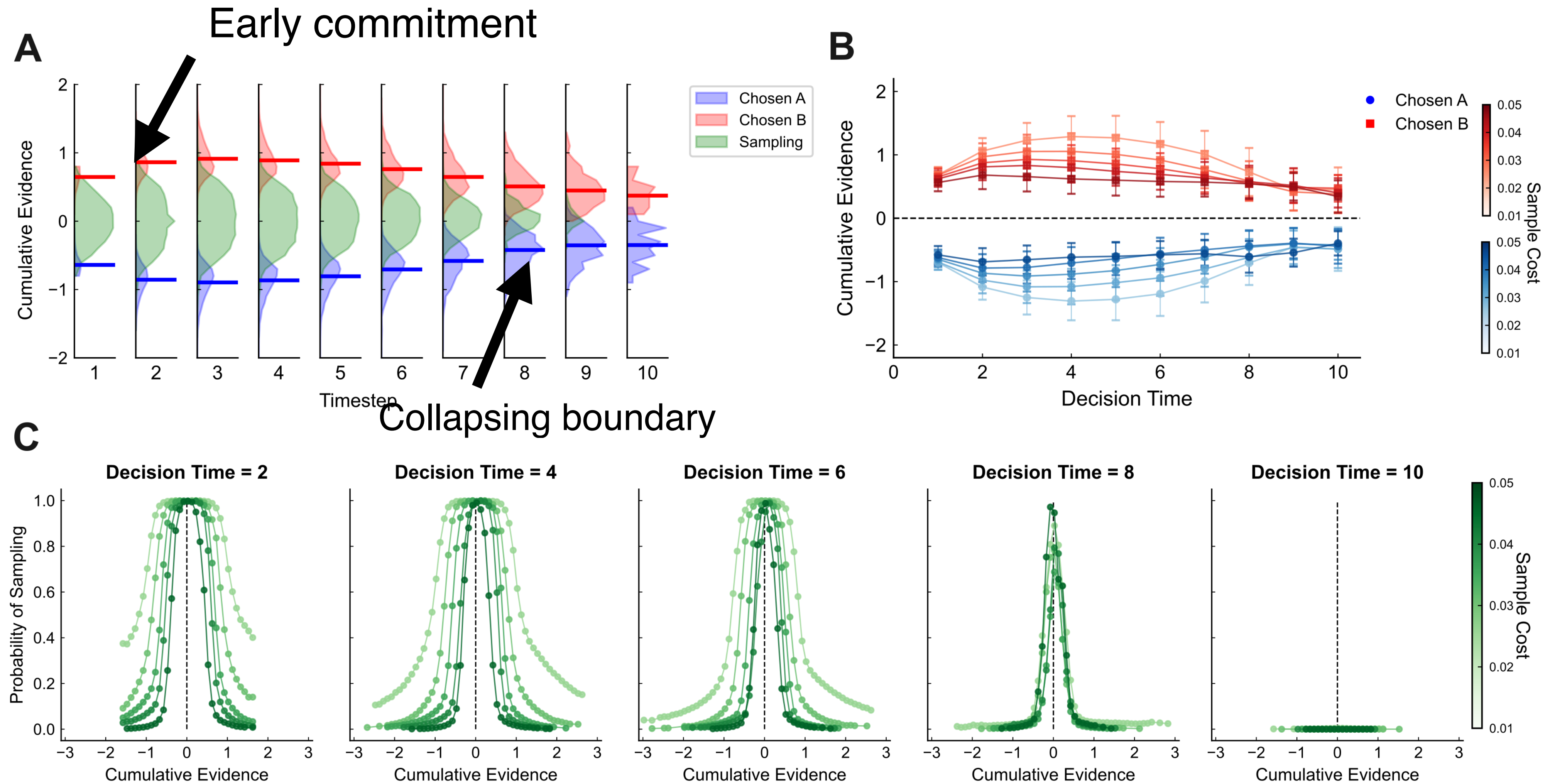
Results

Neural Network Show early commitment behavior at the early stage of decision making



Results

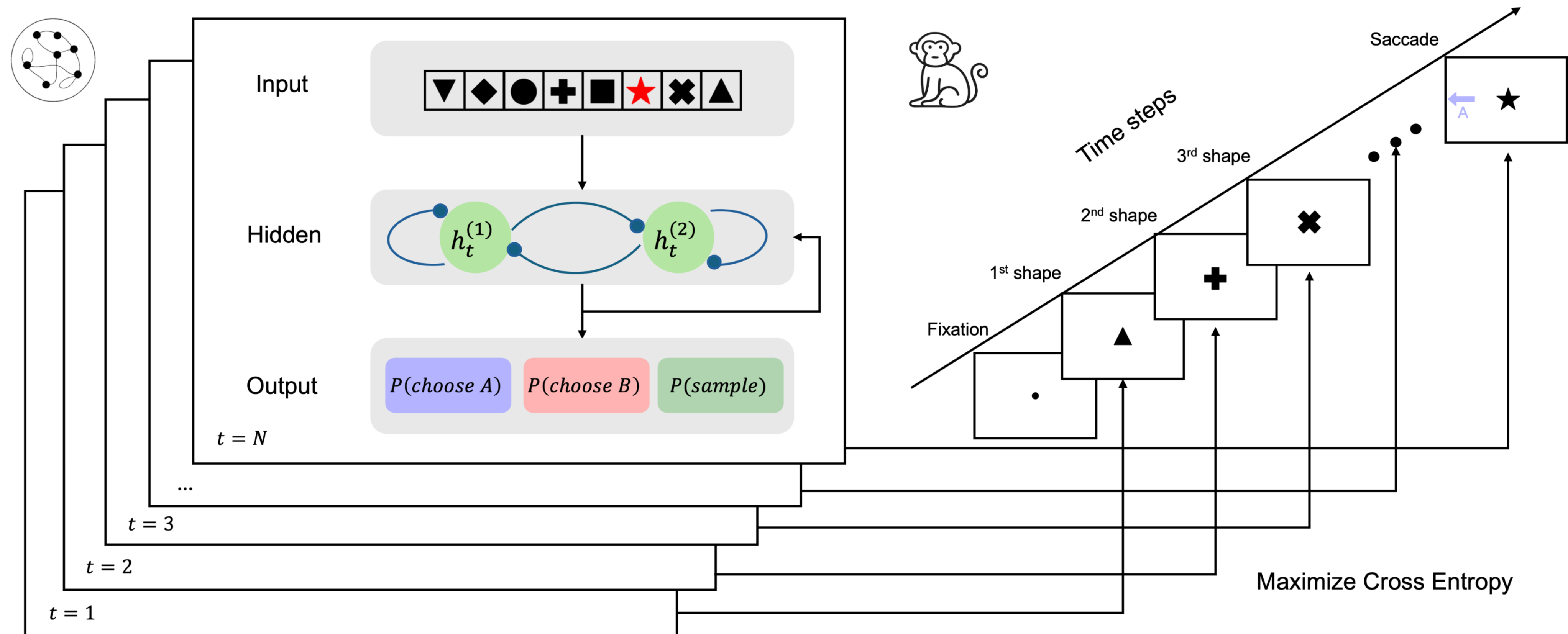
Neural Network learns time-varying decision boundary under time constraints



Methods

Using TinyRNN to discover cognitive strategies in sequential decision making

A

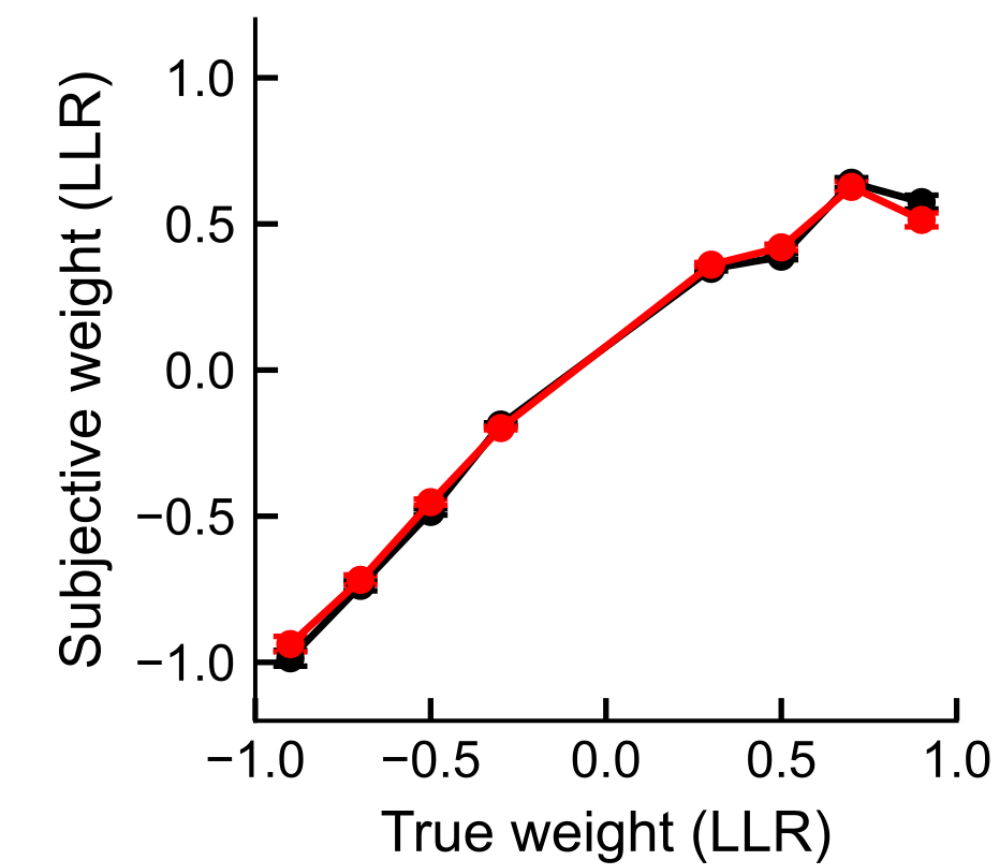
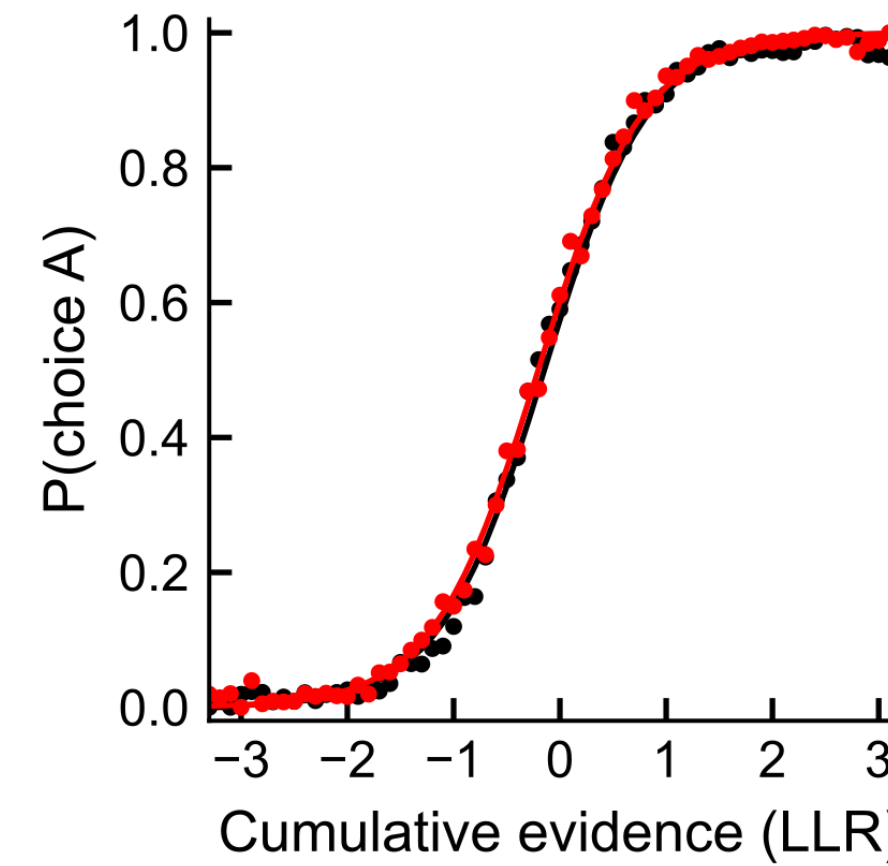
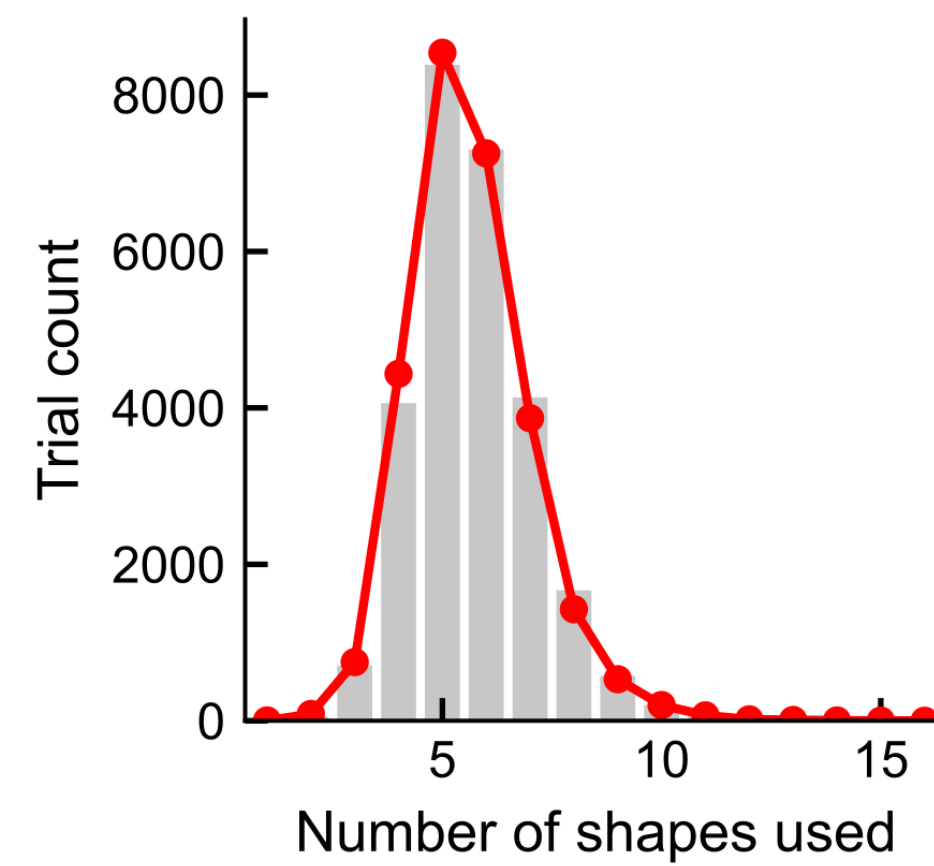


Results

Tiny RNN with only 2 hidden units learn monkey behavioral pattern

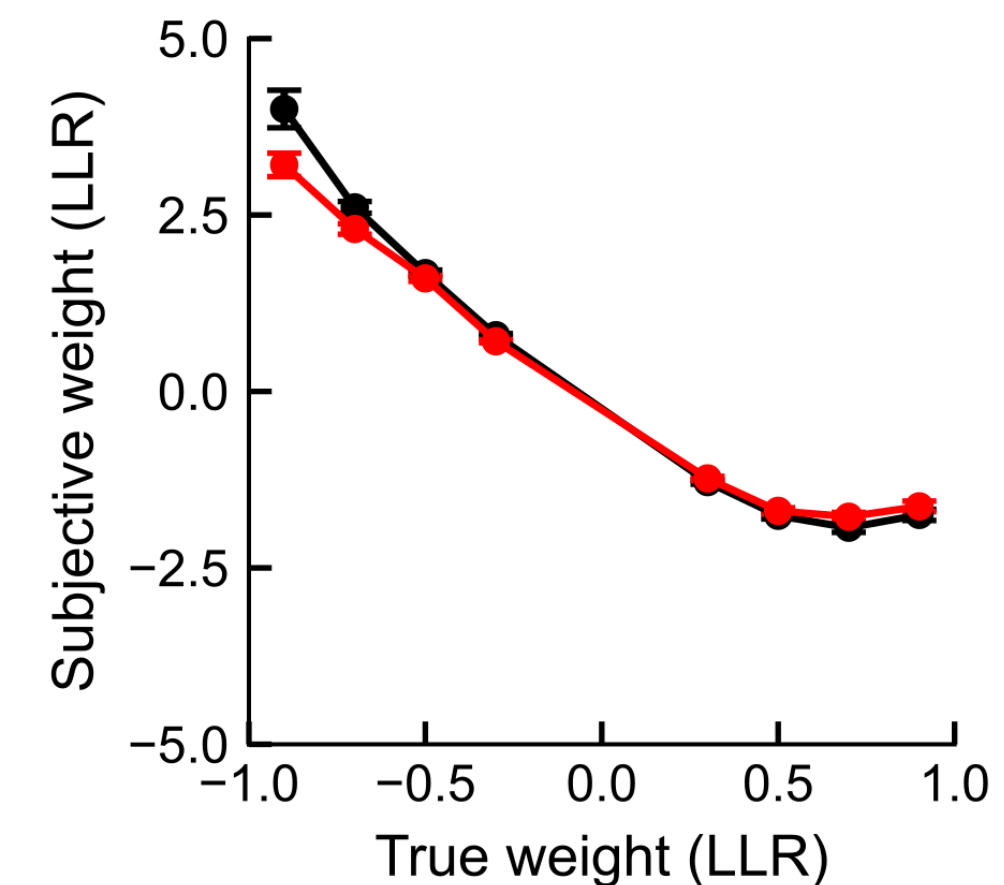
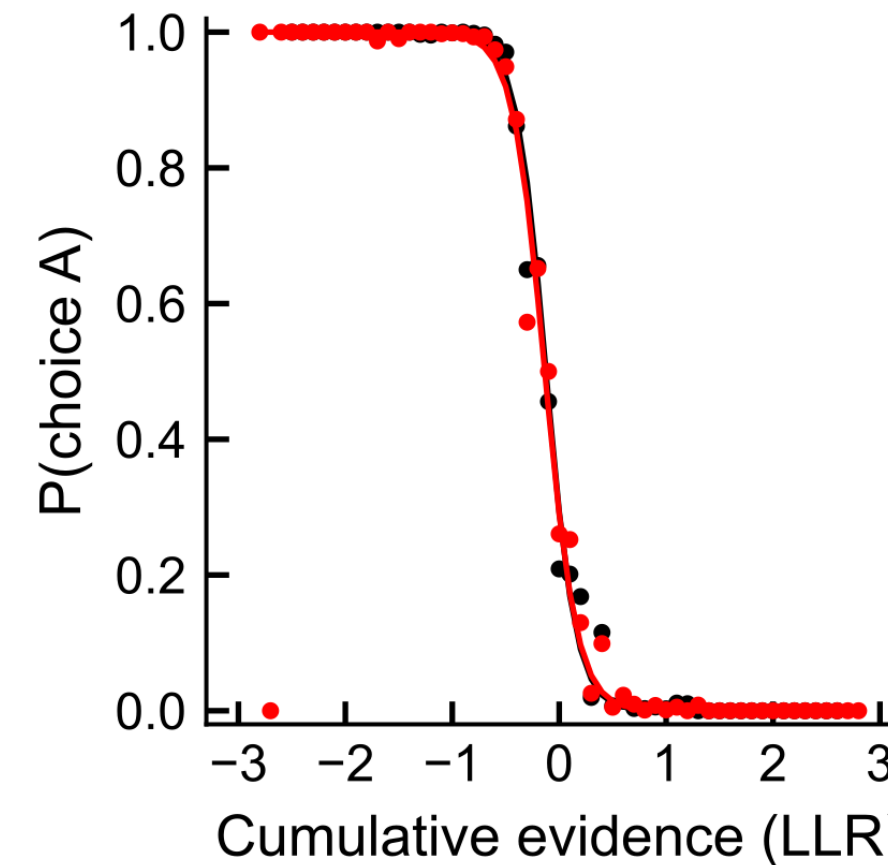
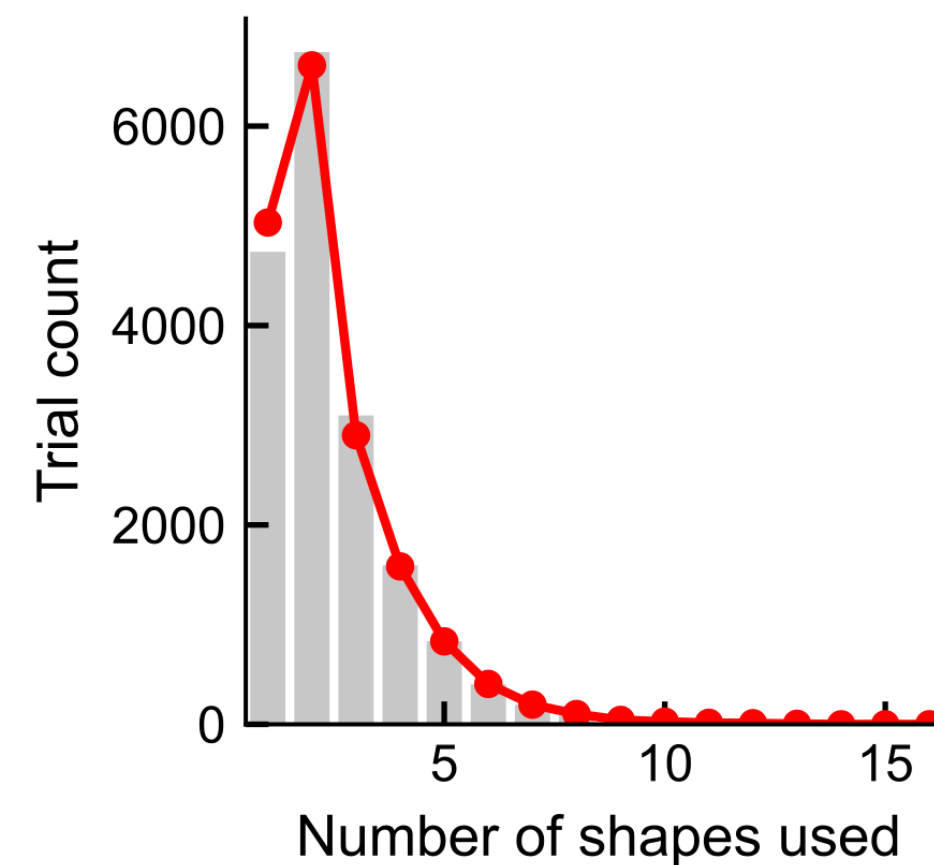
B

Monkey E



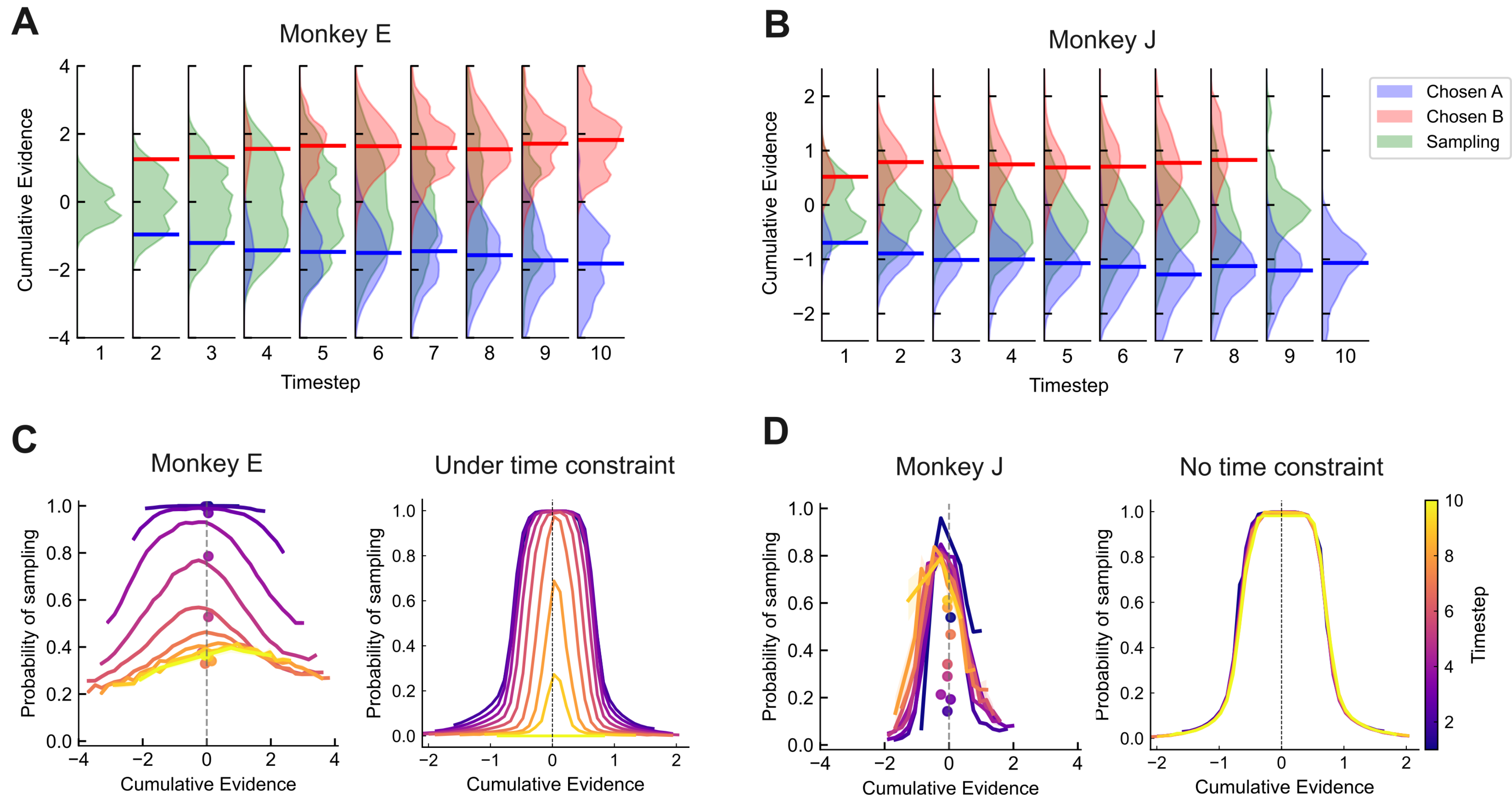
C

Monkey J



Results

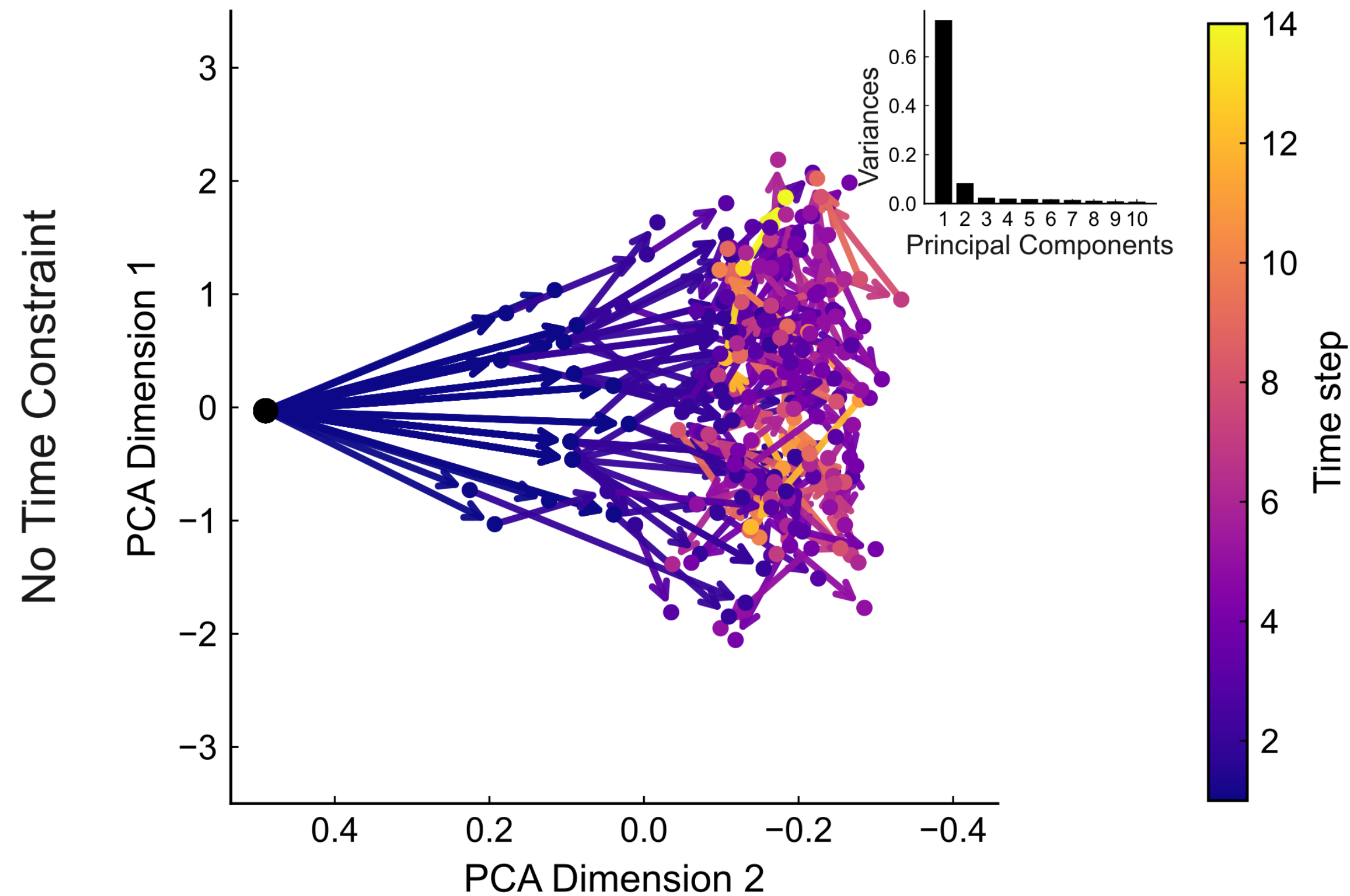
Two monkey strategies are corresponded to task-optimized RNN with/without time constraints



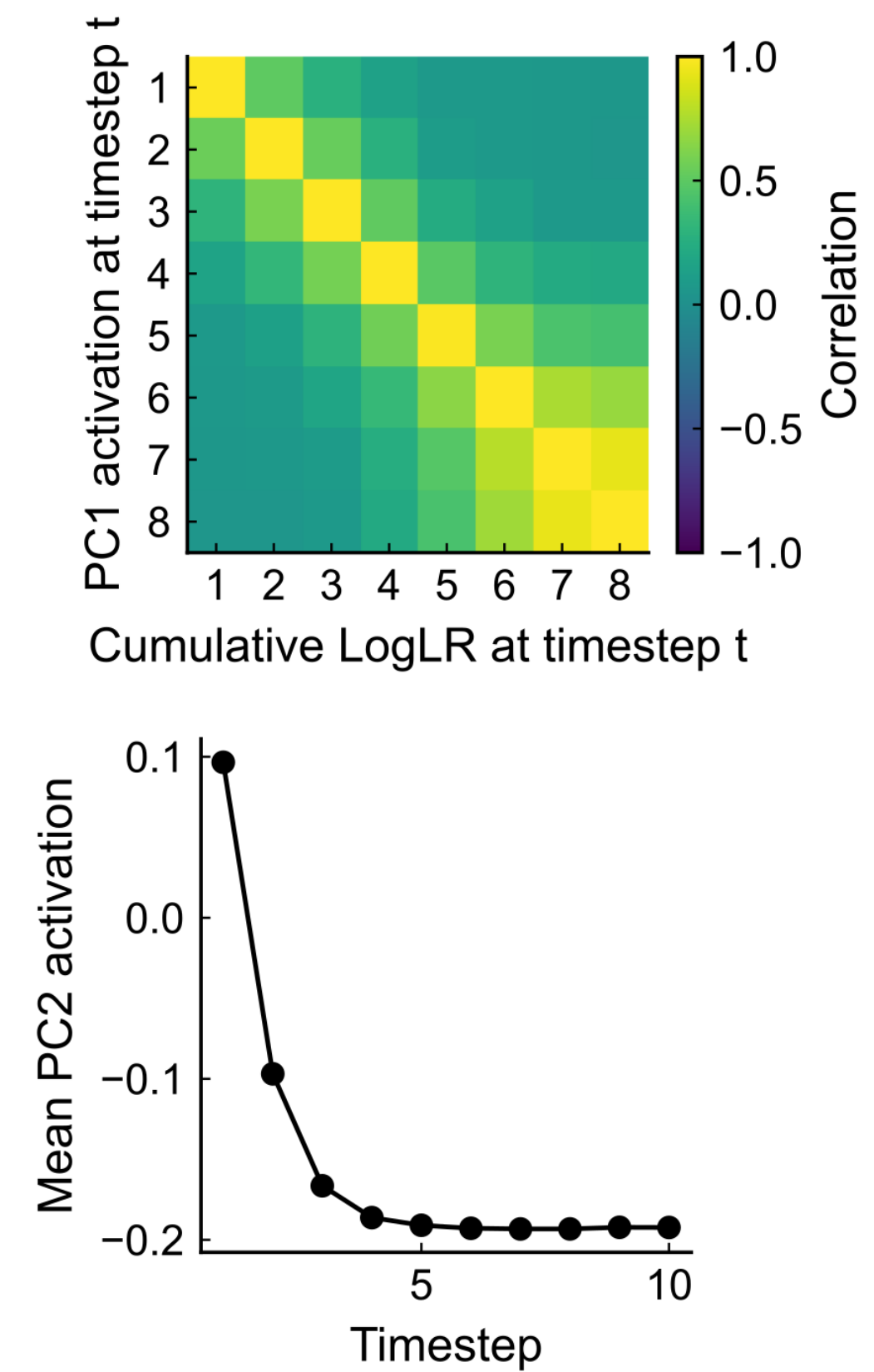
Results

Network distinguishes these early commitment time step in their low dimensional space

A



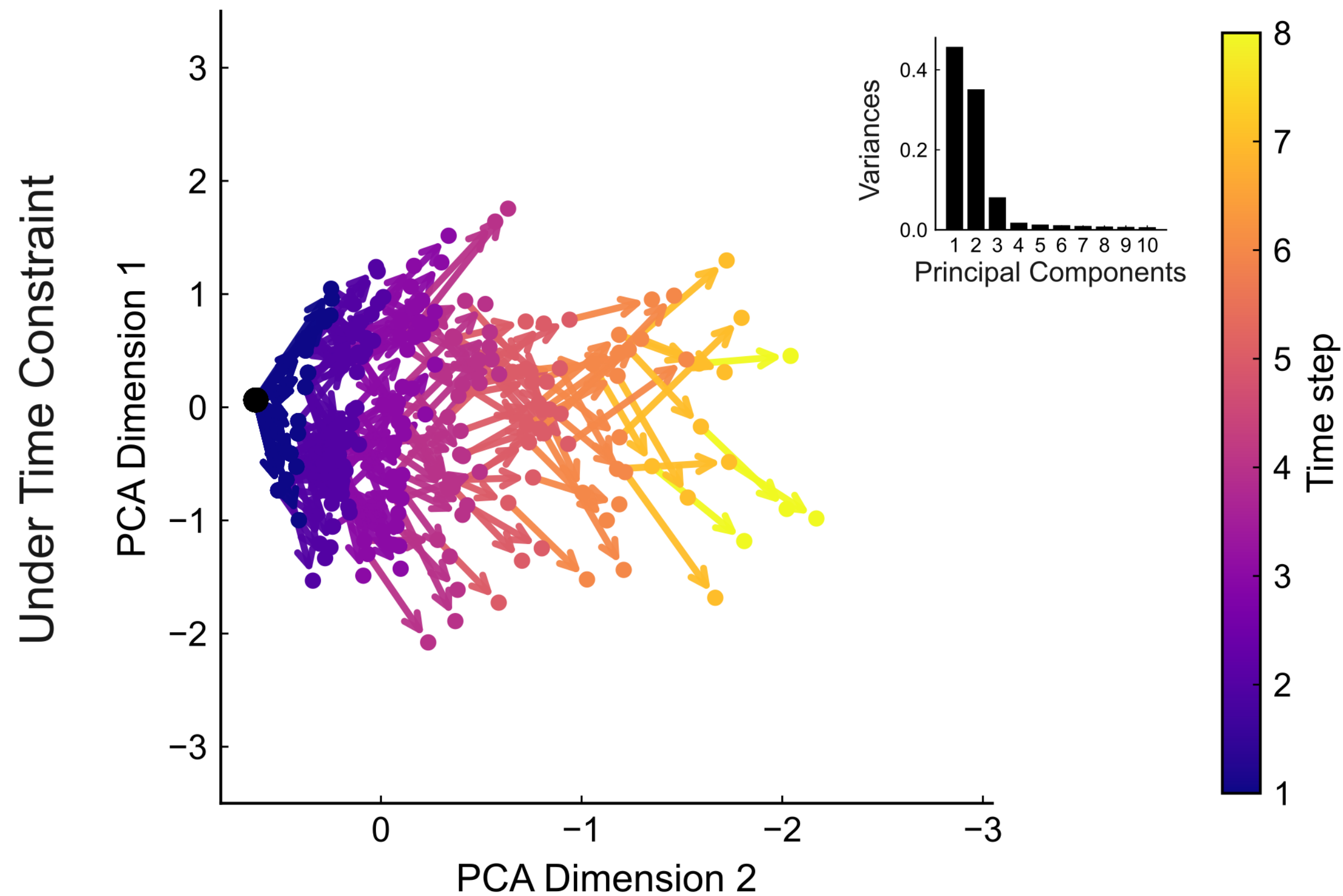
C



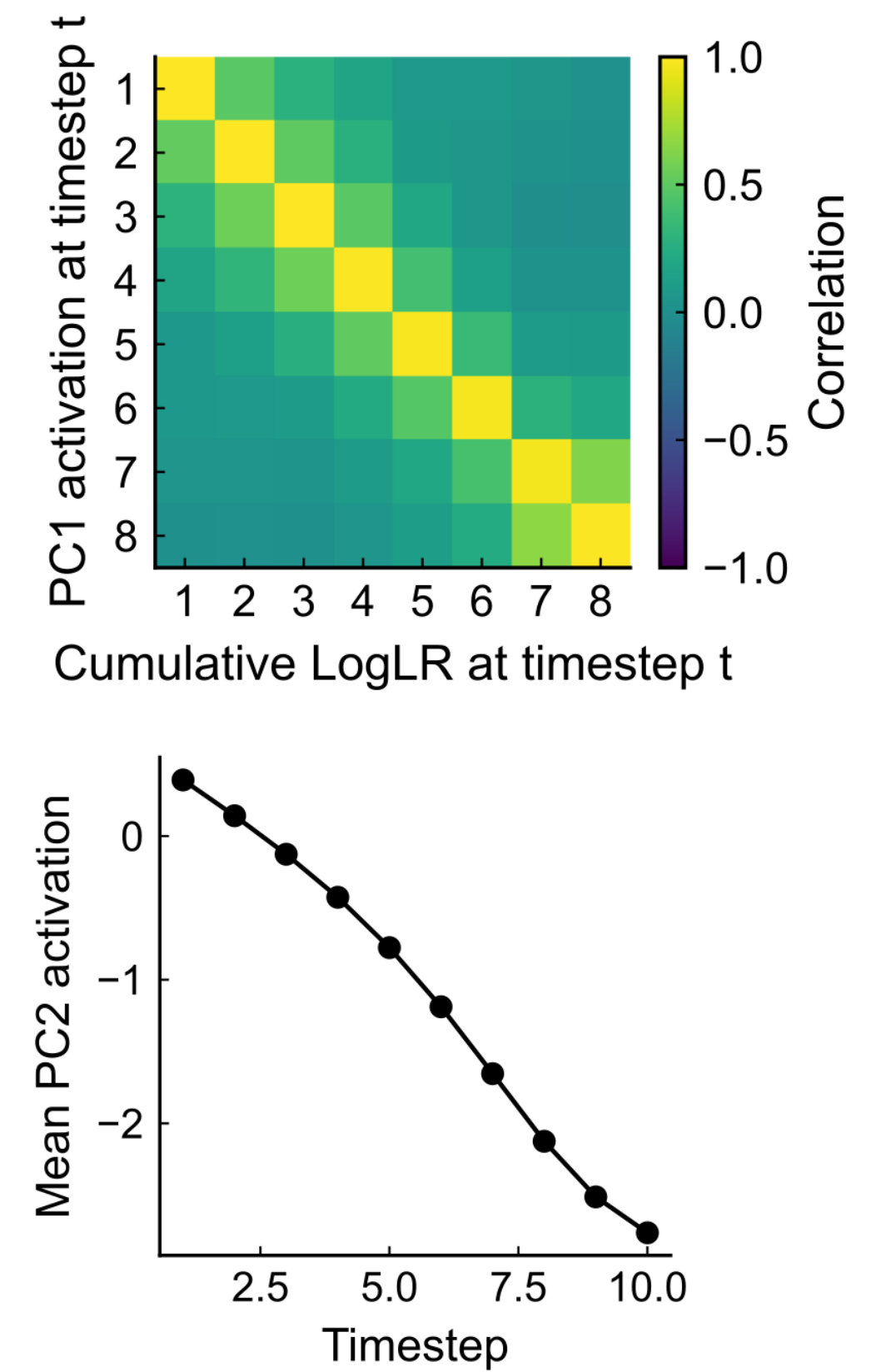
Results

Network distinguishes these early commitment time step in their low dimensional space

B

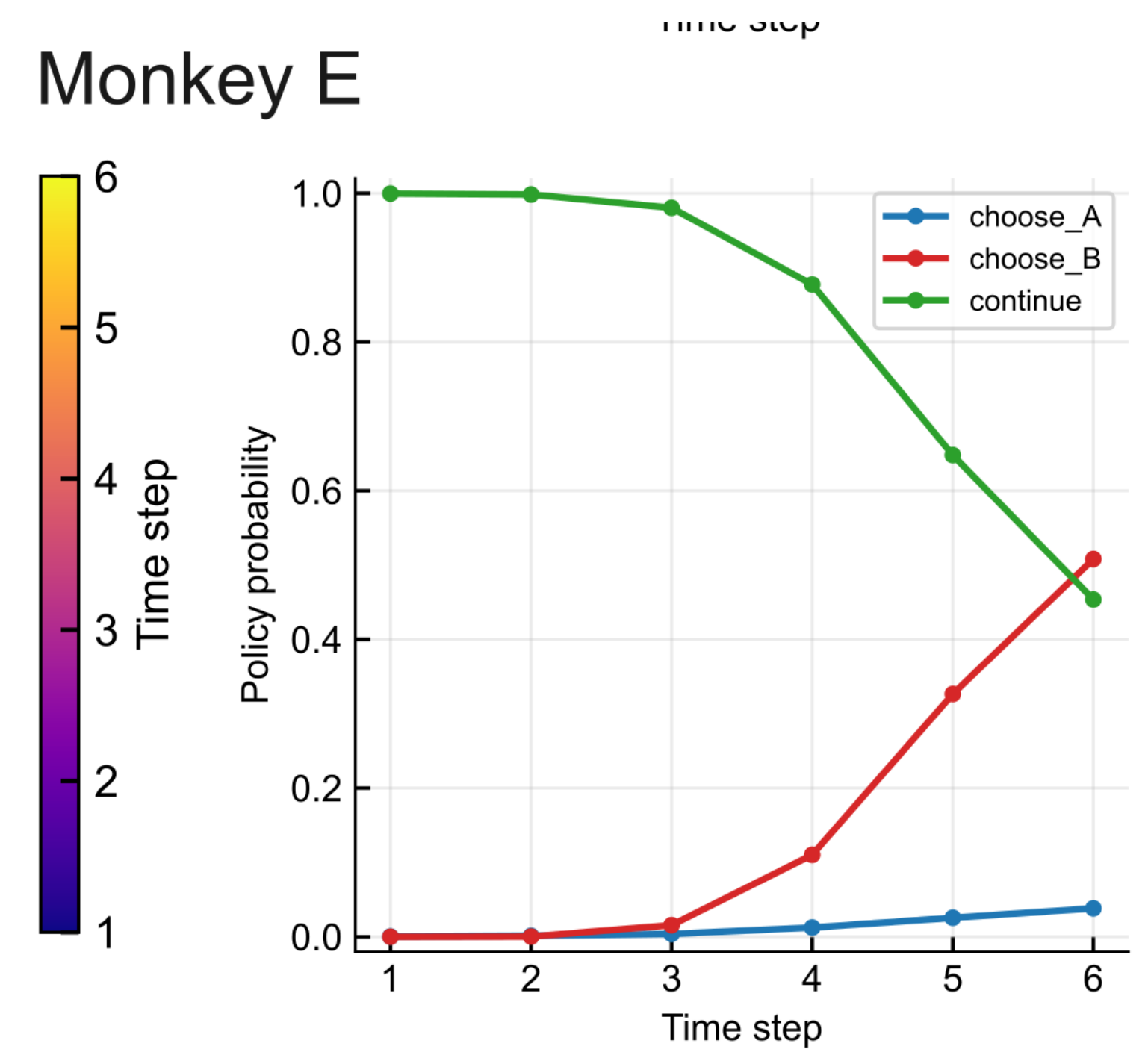
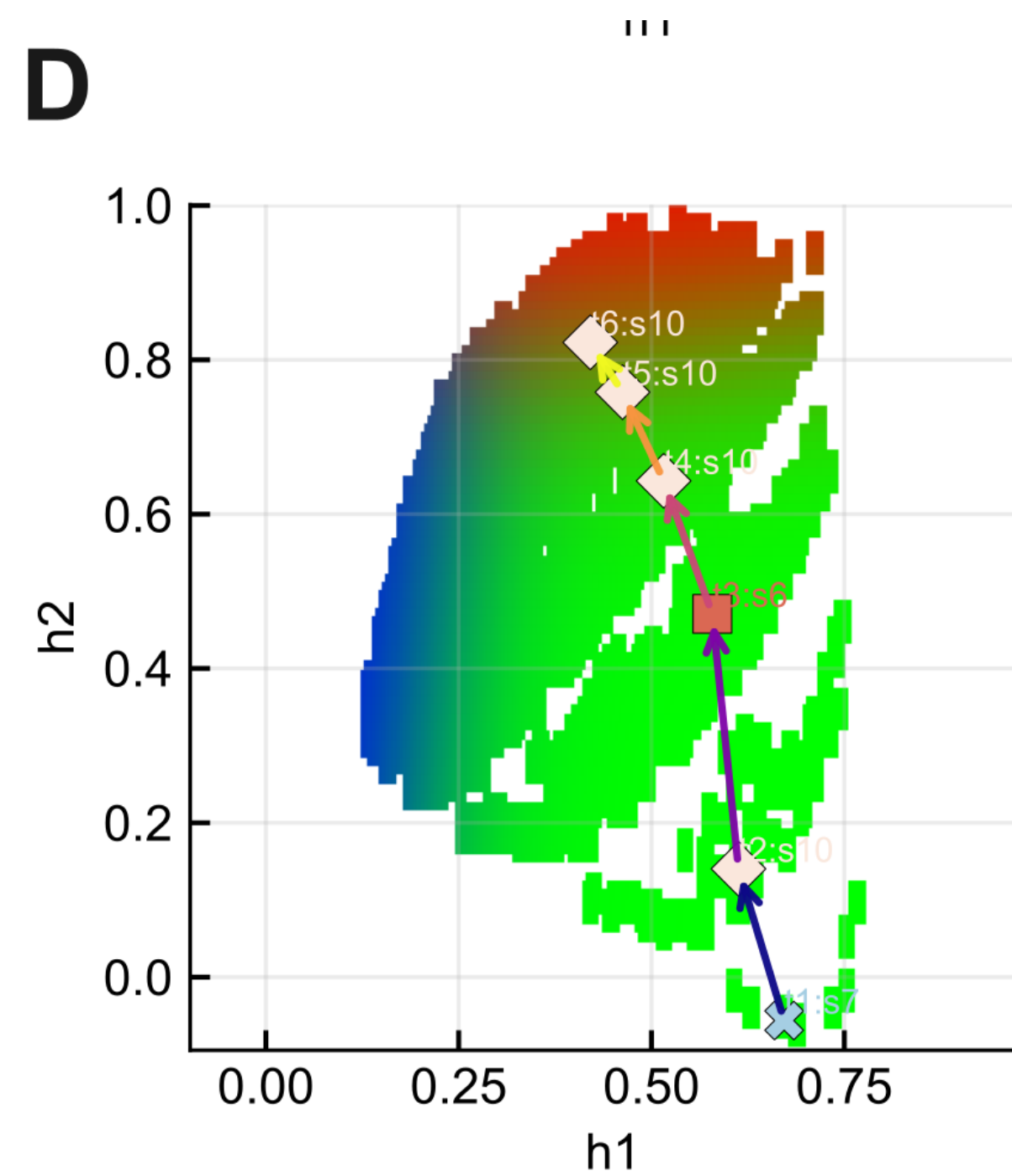
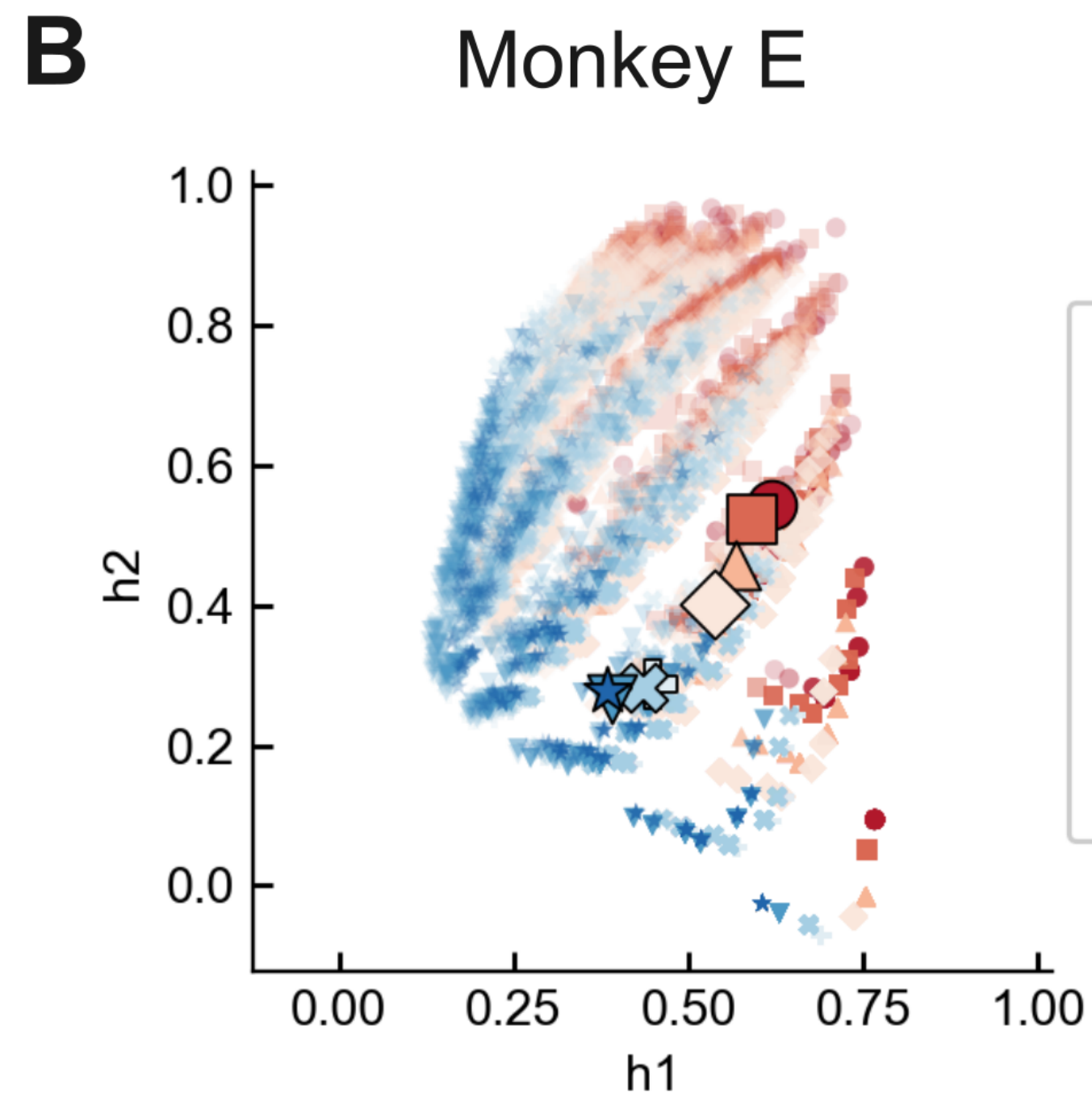


D



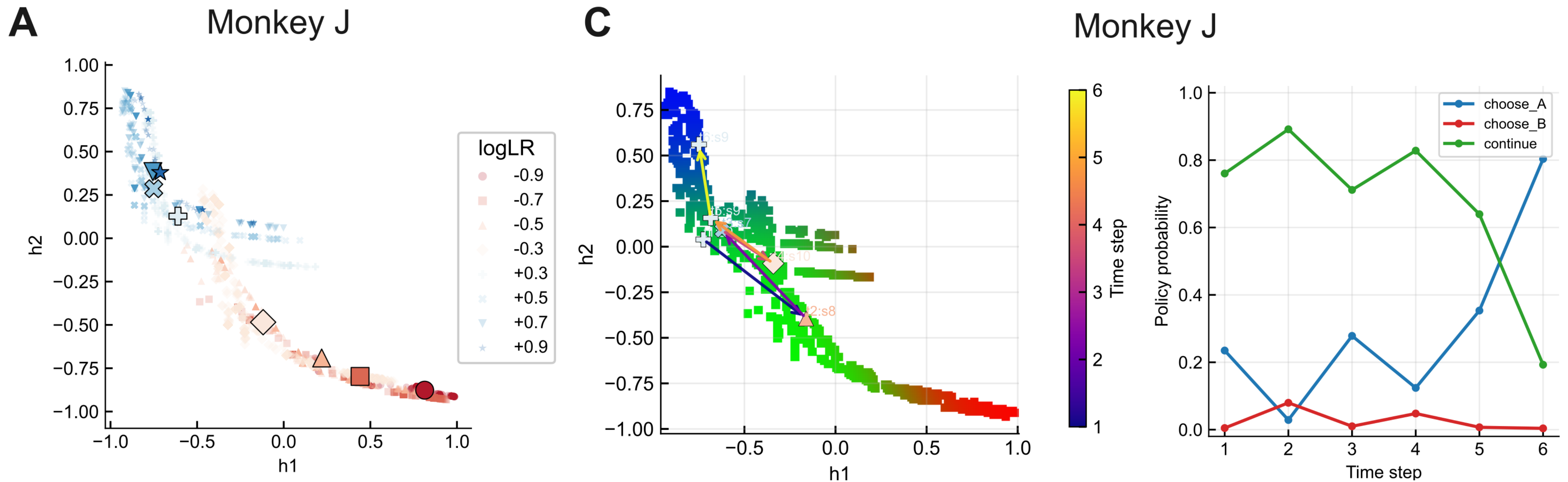
Results

TinyRNN as proxy of monkey exhibit similar representation in hidden states



Results

TinyRNN as proxy of monkey exhibit similar representation in hidden states



Methods

Normative Analysis

At each step, the agent maintains a belief $b_t = P(H_1 | x_{1:t})$ over two hypotheses H_0, H_1 . When agent receive a new sample, it update the posterior through:

$$b_t = p(H_1 | x_{1:t}) = \frac{p(x_t | H_1) \cdot b_{t-1}}{p(x_t)} = \frac{f_1(x_t) \cdot b_{t-1}}{f_1(x_t) \cdot b_{t-1} + f_0(x_t) \cdot (1 - b_{t-1})}$$

To derive the optimal policy, the bellman optimality is used to calculate the value of each action:

$$V^*(b, t) = \max \left\{ \underbrace{R_0(b_t, \perp)}_{\text{Choose } H_0}, \underbrace{R_1(b_t, \perp)}_{\text{Choose } H_1}, \underbrace{-c + \mathbb{E}_{x|b} [V(b'(x))]}_{\text{Sample}} \right\}$$

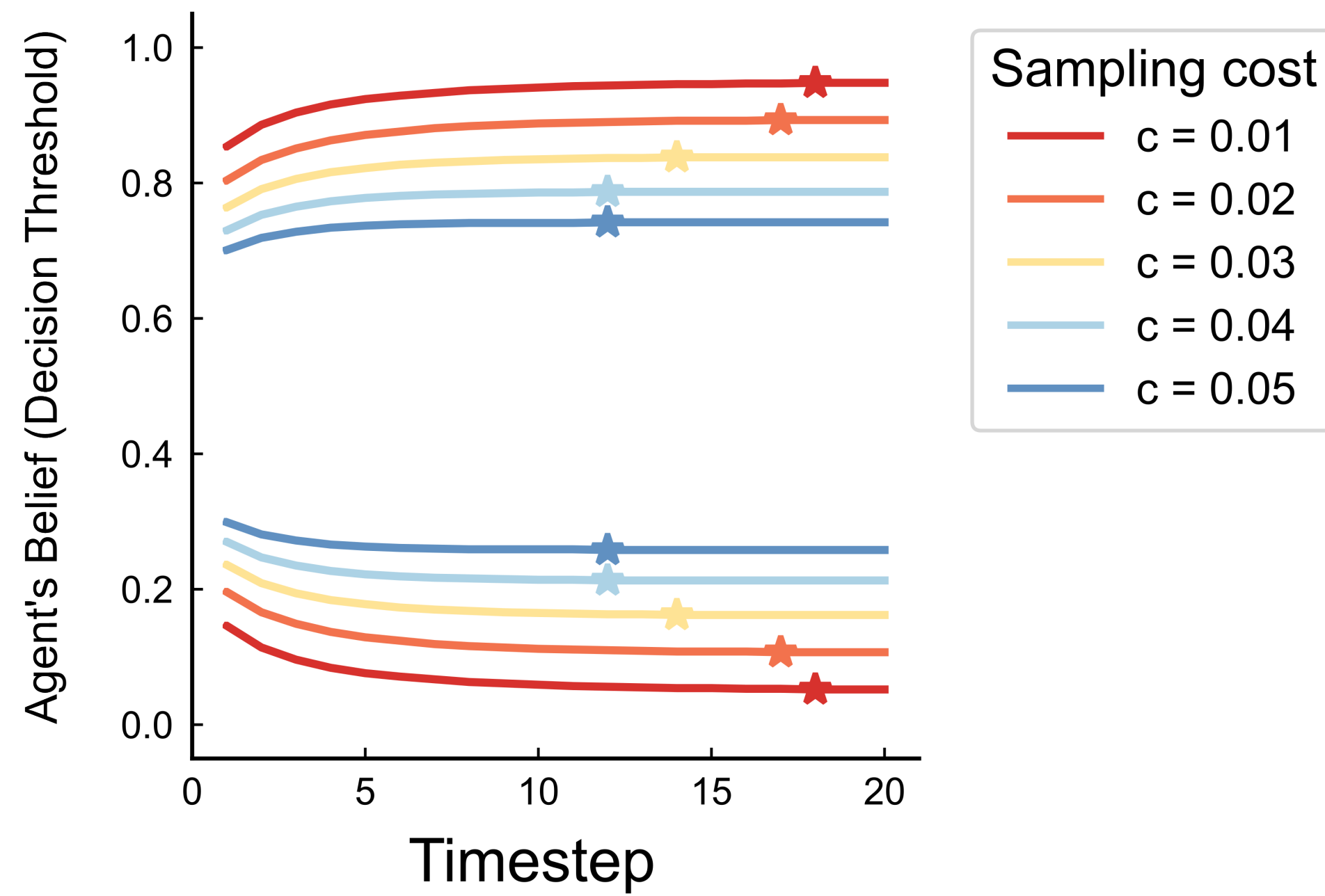
The value of sampling is given by iterating over every possible belief in the next time step $t+1$:

$$\mathbb{E}_{x|b} [V_{t+\Delta t} (b'_{t+\Delta t}(x))] = \int_{-\infty}^{\infty} V_{t+\Delta t} (b'_{t+\Delta t}(x)) f(x | b) dx$$

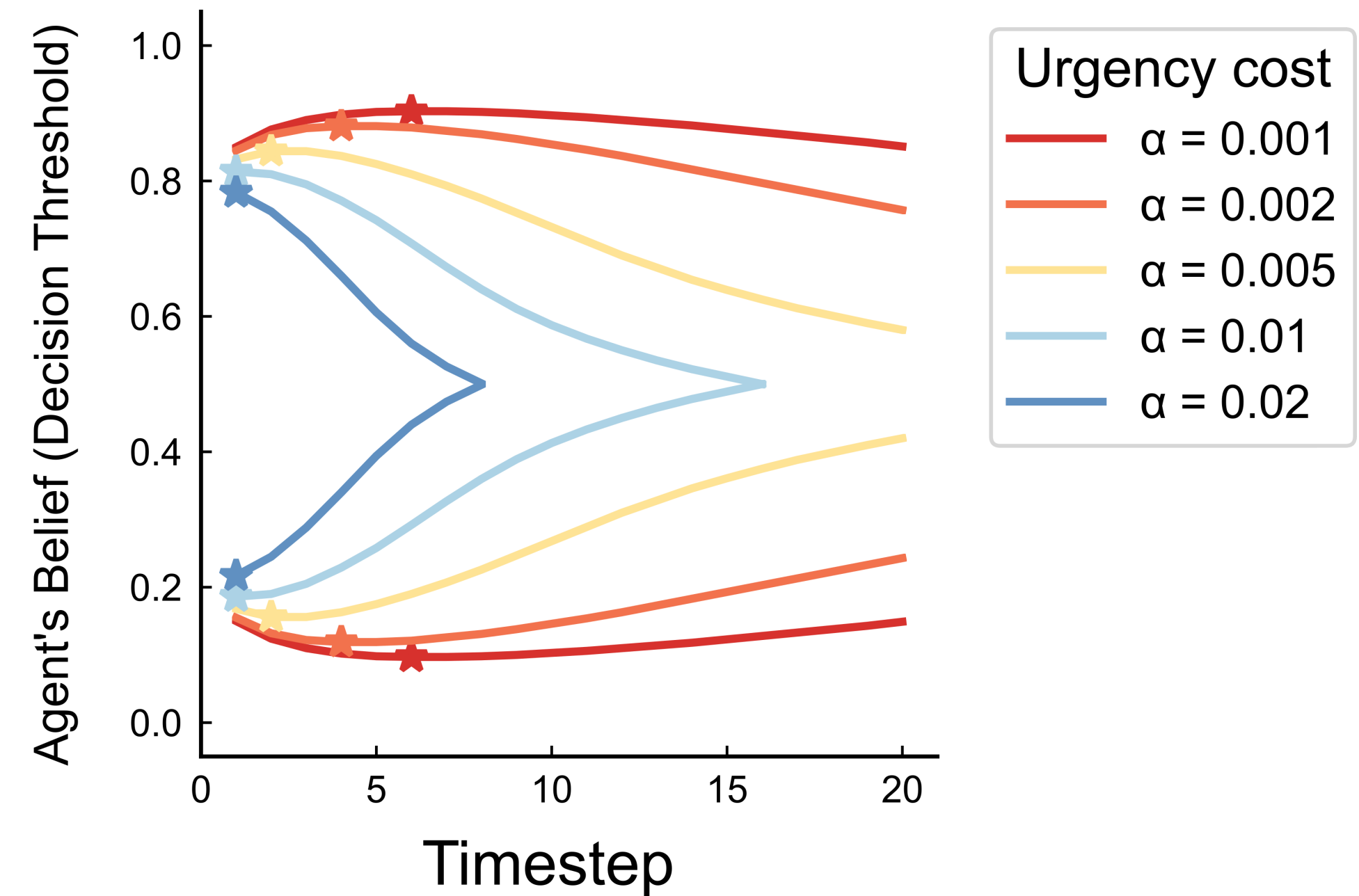
Results

Normative Analysis

No Time Constraint



Under Time Constraint



Conclusion

Novel Optimal Stopping Rule in Heterogeneous Environment

- We identified an early commitment as a novel stopping strategy other than the well-known collapsing boundary in heterogeneous environment.
- We proved early commitment as an optimal stopping rule in heterogeneous evidence stream environment using dynamic programming method.